

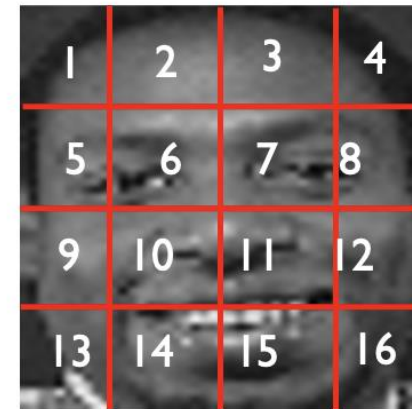


Extracción de Características HoG + LBP

(CAPÍTULO 2 – PR02D)

Domingo Mery 

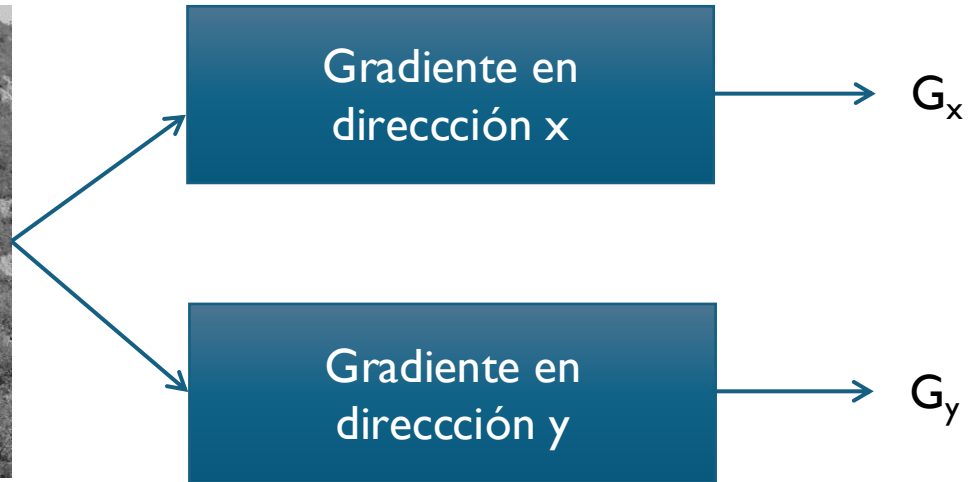
Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>



Características de Intensidad

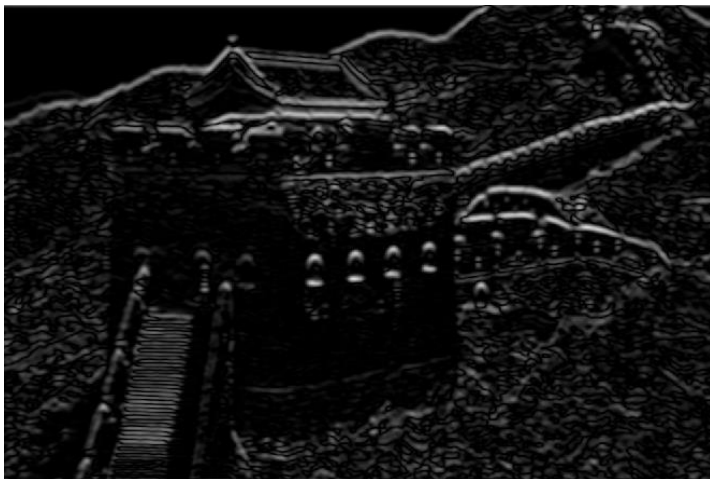
Histogramas de Gradiente

Histograma de Gradients

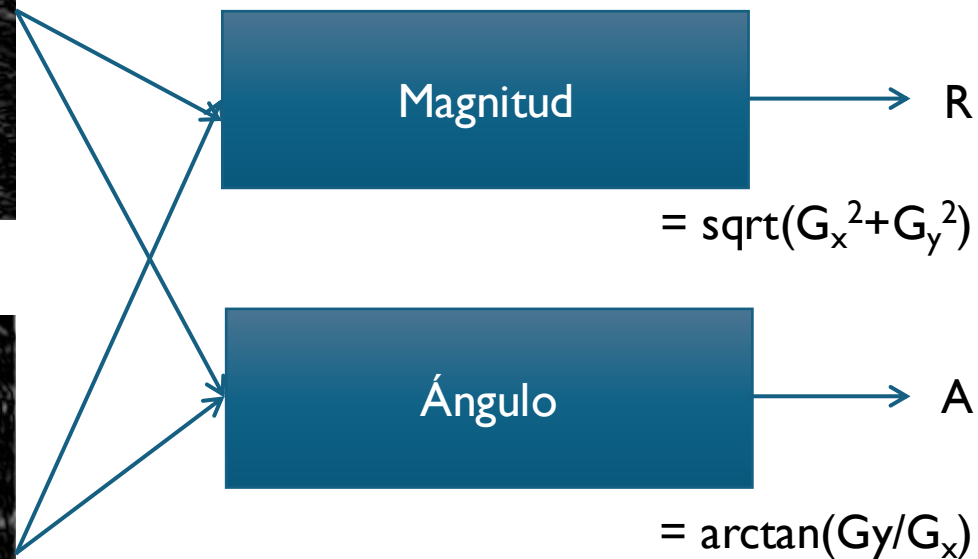


Histogram of Gradients

G_x : Gradiente en dirección x



G_y : Gradiente en dirección y

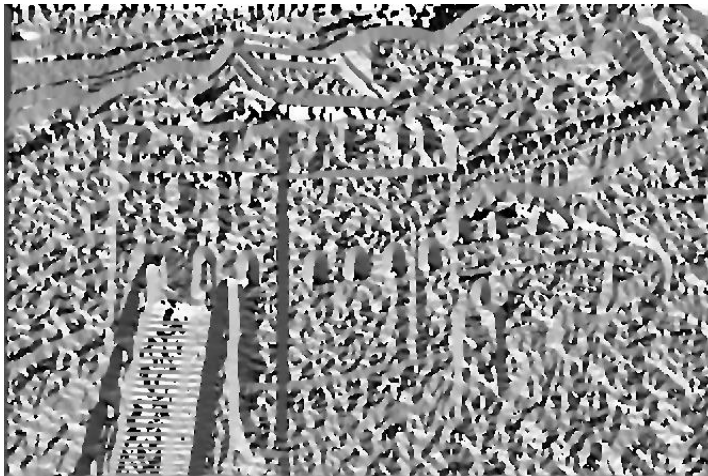


Histograma de Gradientes

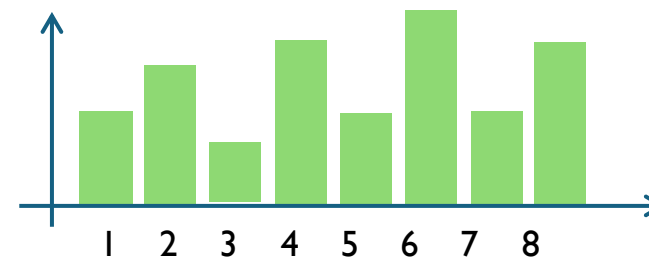
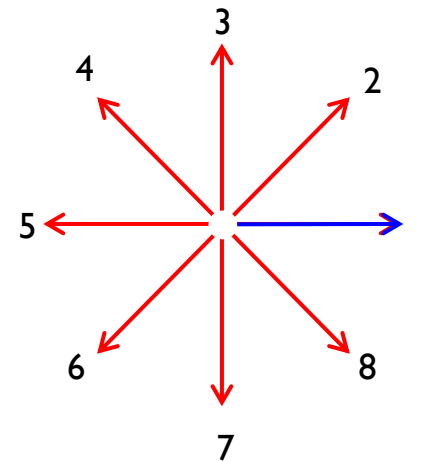
R: Magnitud



A: Ángulo



Histograma de 8 direcciones

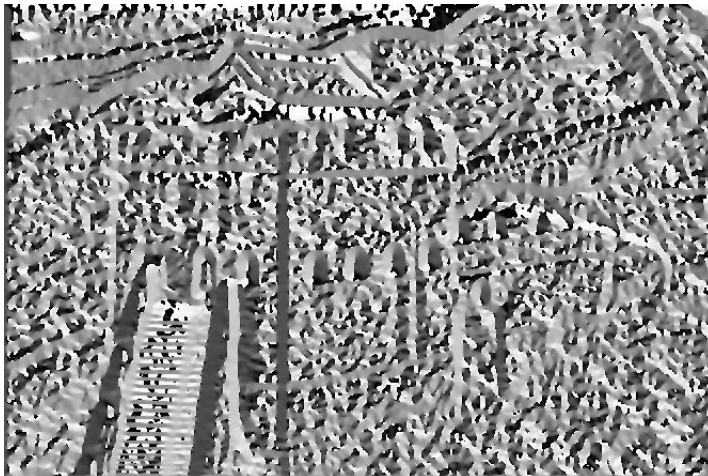


Histograma de Gradientes

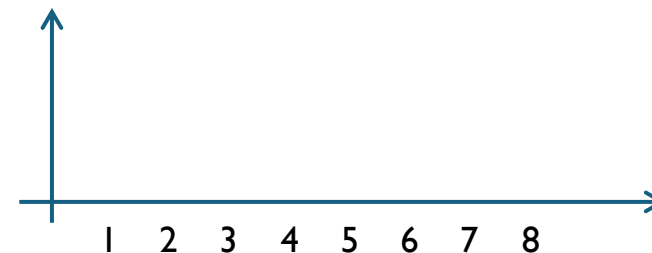
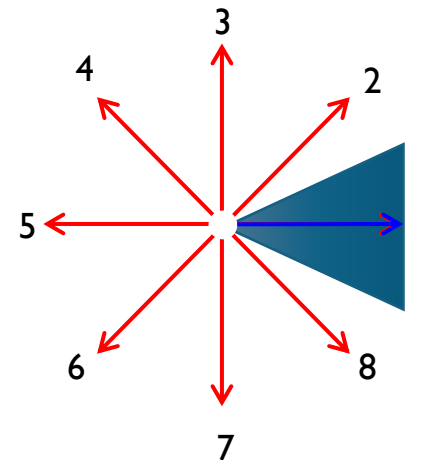
R: Magnitud



A: Ángulo



Histograma de 8 direcciones
(cálculo del primer 'bin')



Histograma de Gradientes

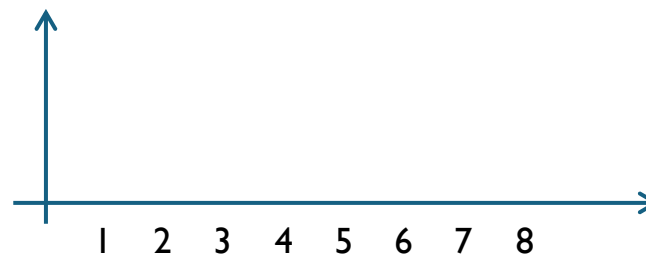
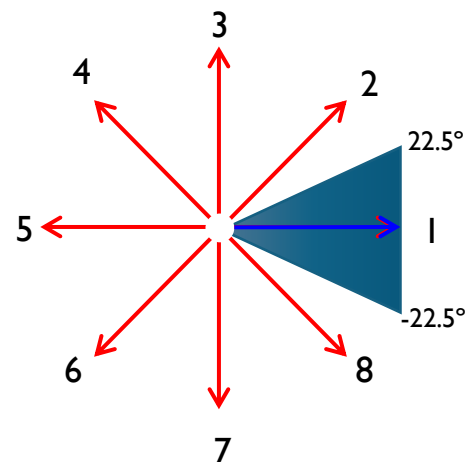
R: Magnitud



A: Ángulo between -22.5° and 22.5°

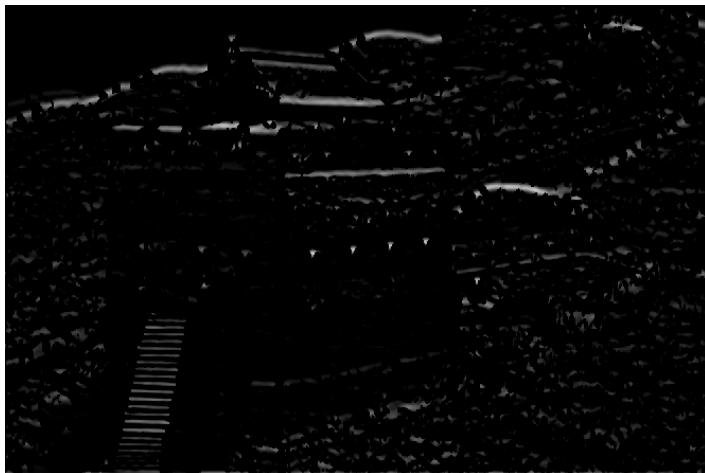


Histograma de 8 direcciones
(cálculo del primer 'bin')



Histograma de Gradientes

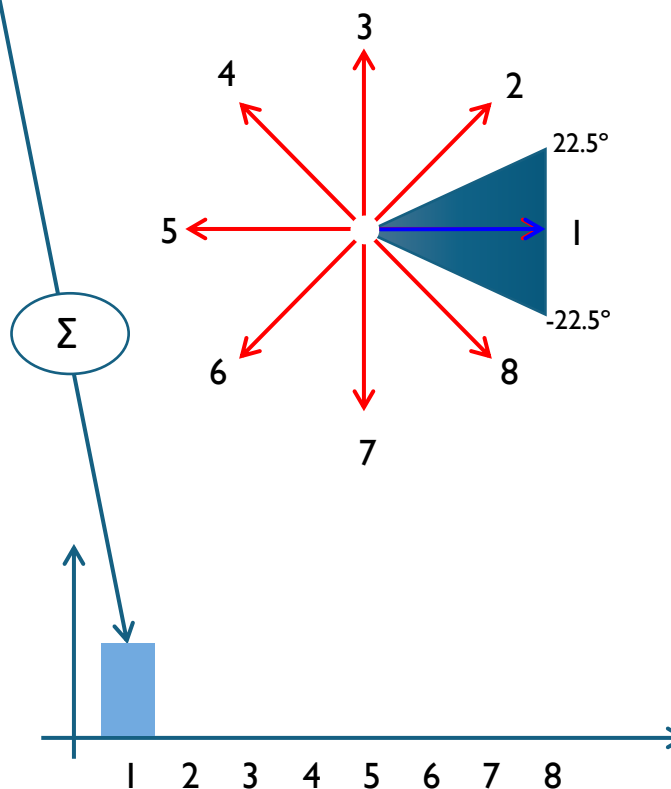
R: Magnitud en esta dirección



A: Ángulo entre -22.5° y 22.5°

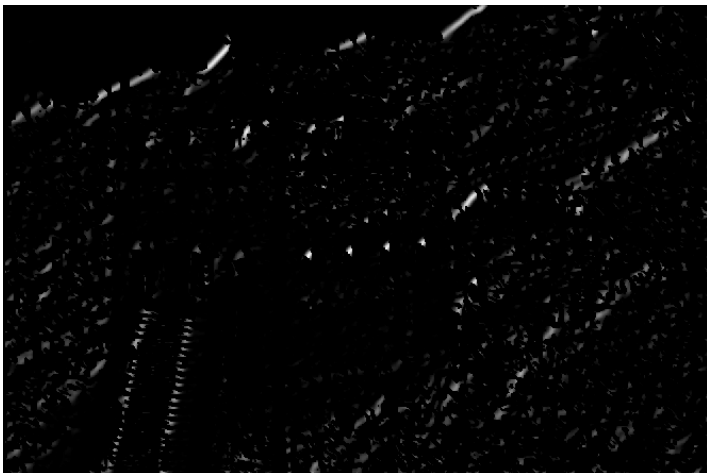


Histograma de 8 direcciones
(cálculo del primer 'bin')



Histograma de Gradientes

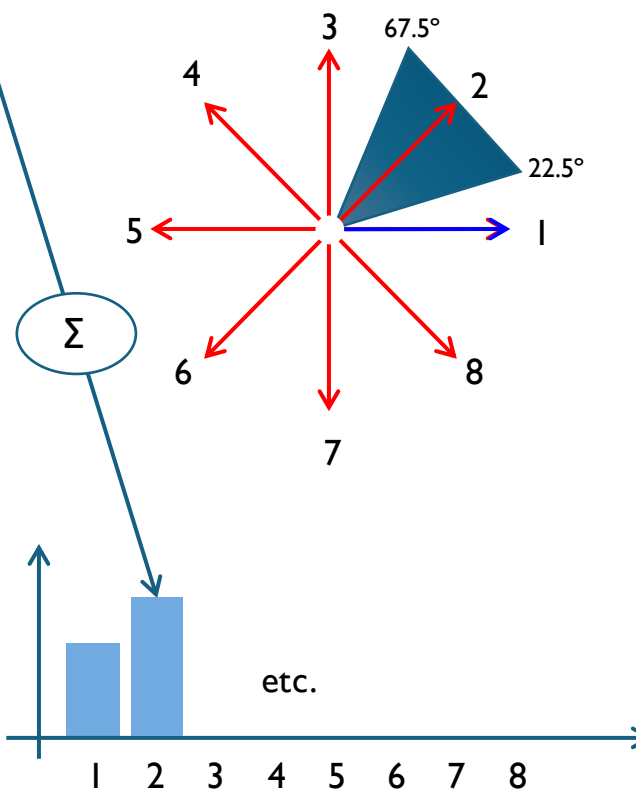
R: Magnitud en esta dirección



A: Ángulo entre 22.5° y 67.5°



Histograma de 8 direcciones
(cálculo del segundo 'bin')

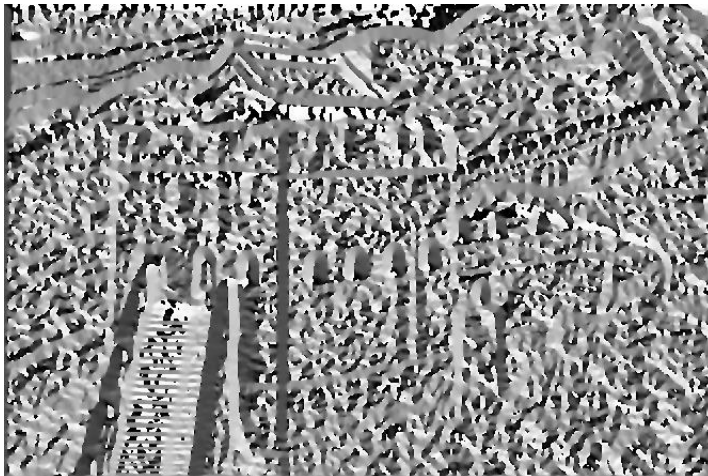


Histograma de Gradientes

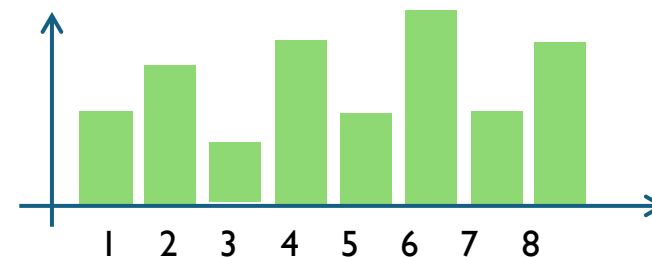
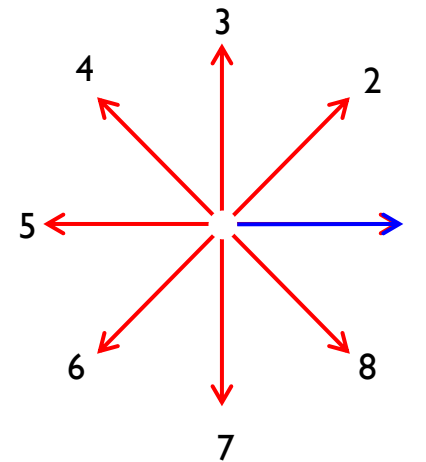
R: Magnitud



A: Ángulo



Histograma de 8 direcciones

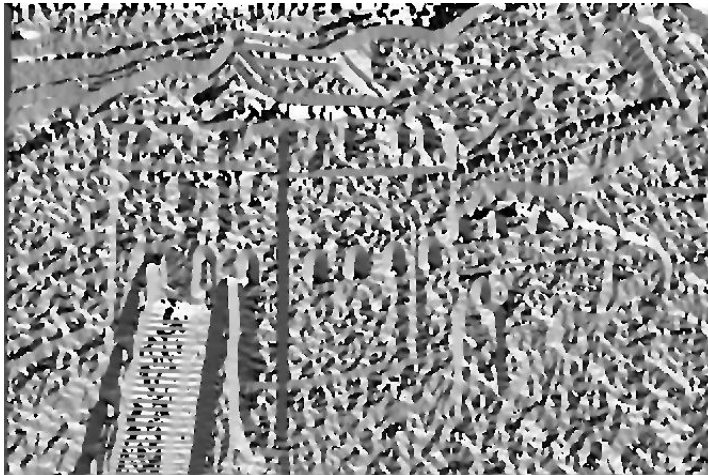


Histograma de Gradientes

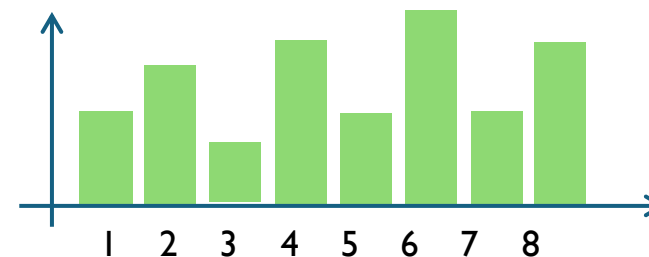
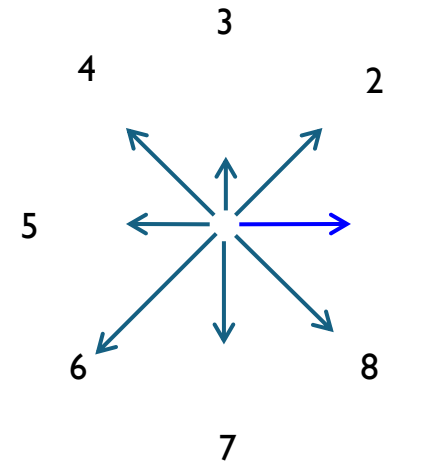
R: Magnitud



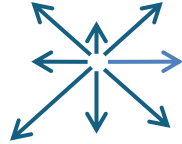
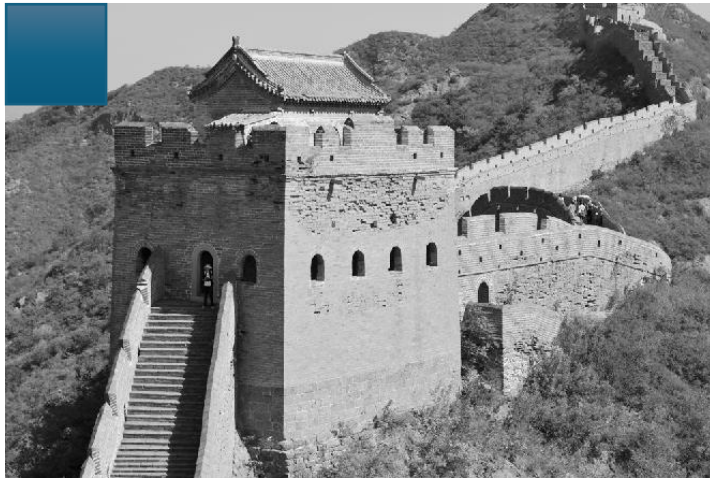
A: Ángulo

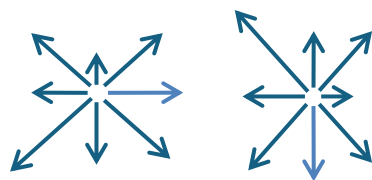


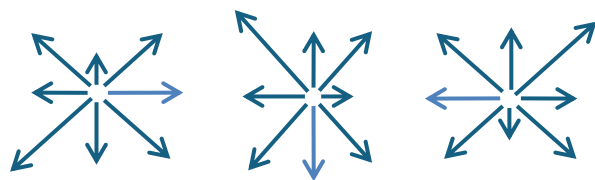
Histograma de 8 direcciones

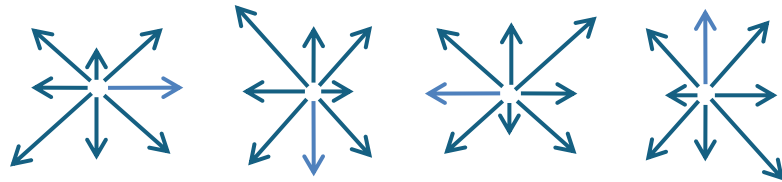


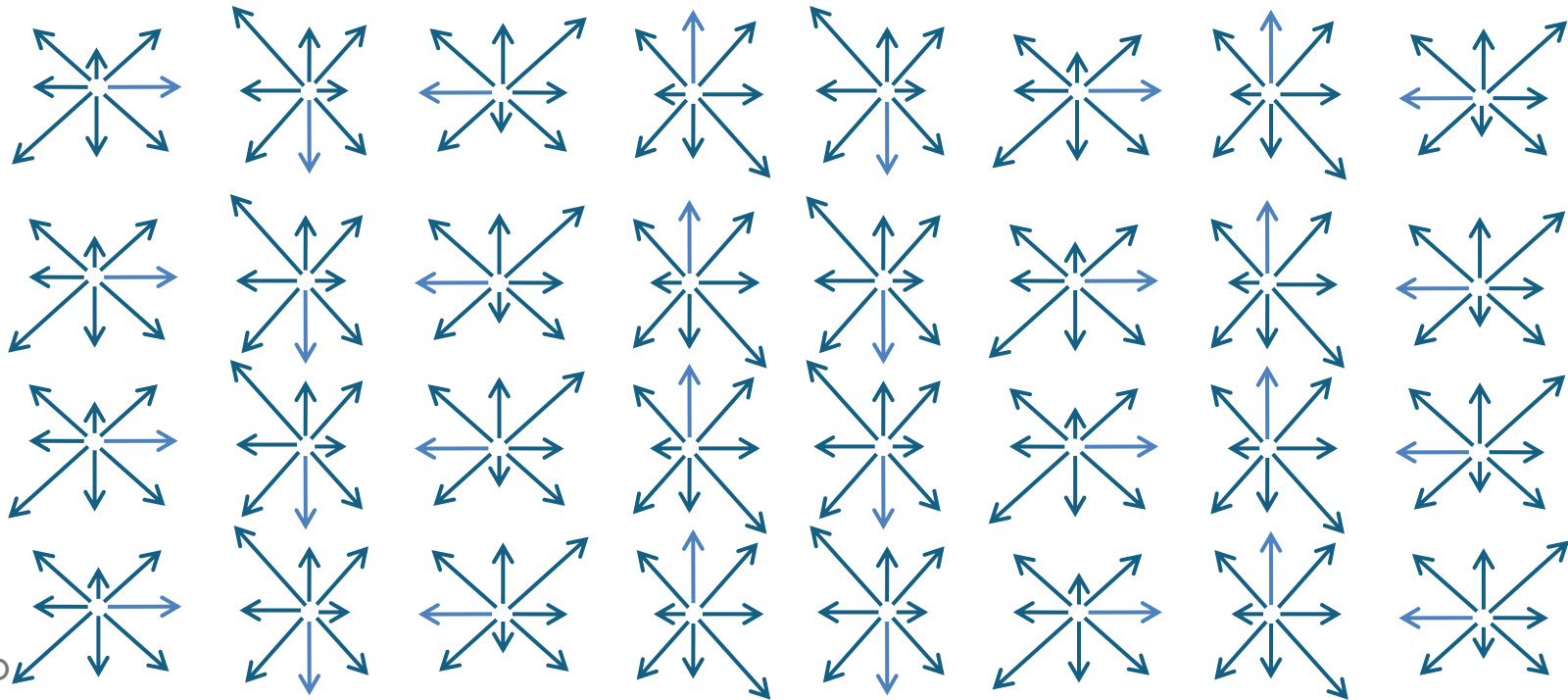
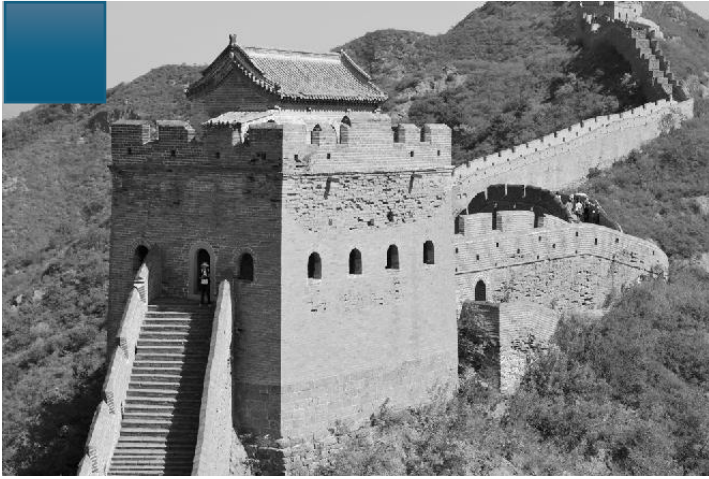
El descriptor propuesto por los autores es una concatenación de HoG en diferentes particiones superpuestas

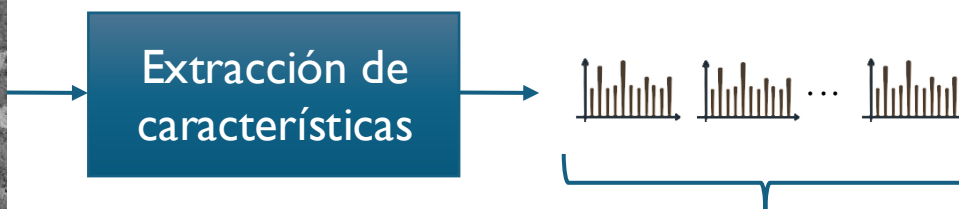




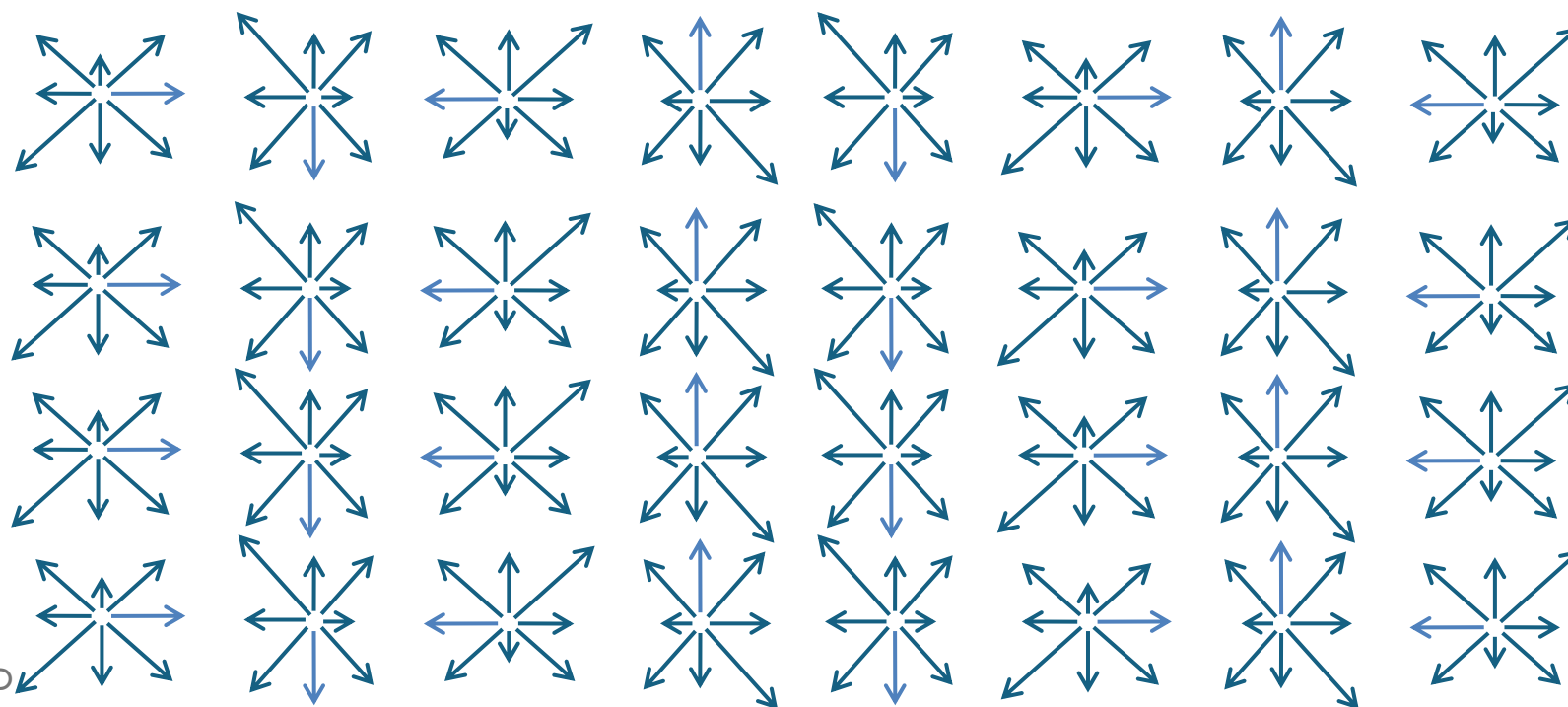




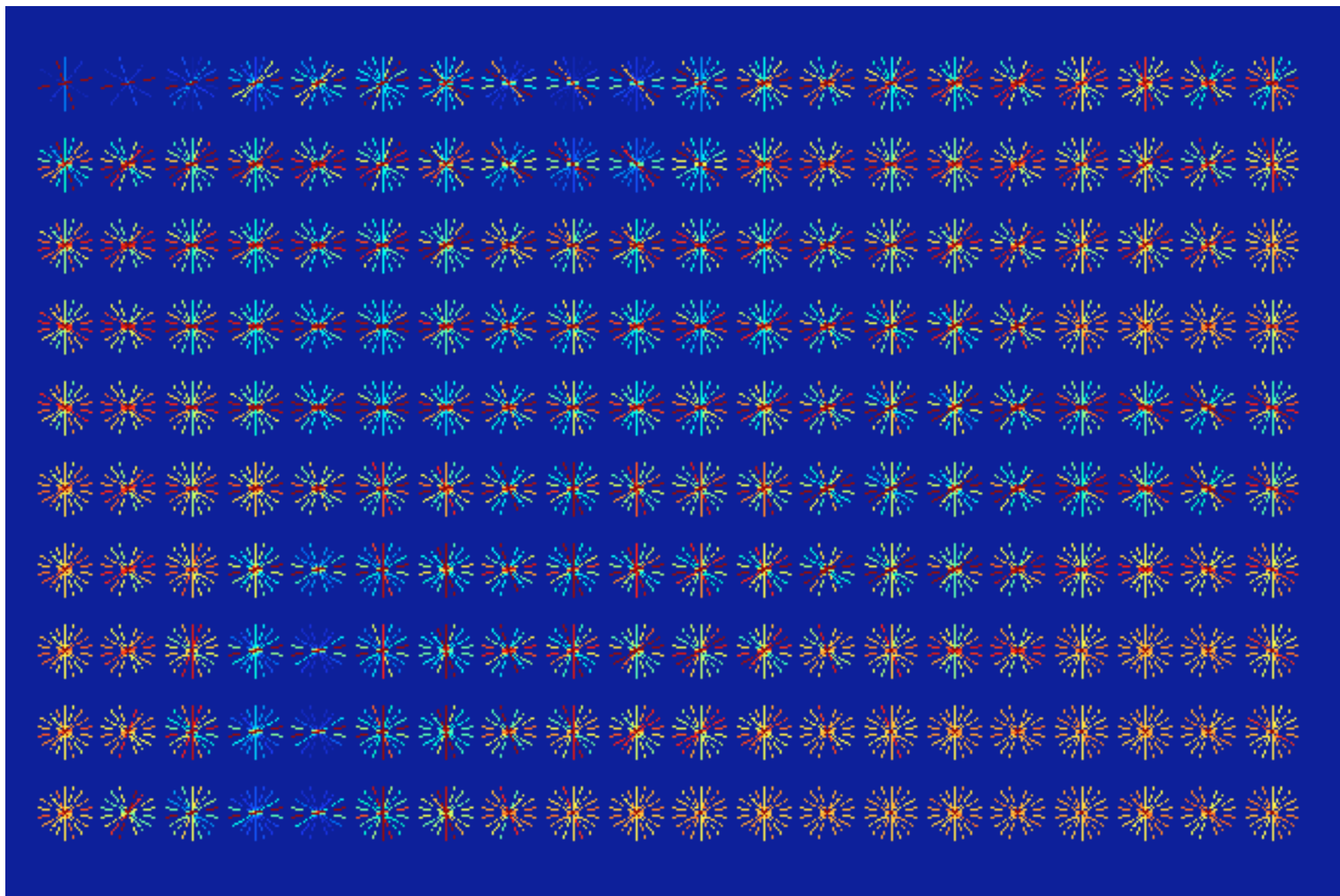




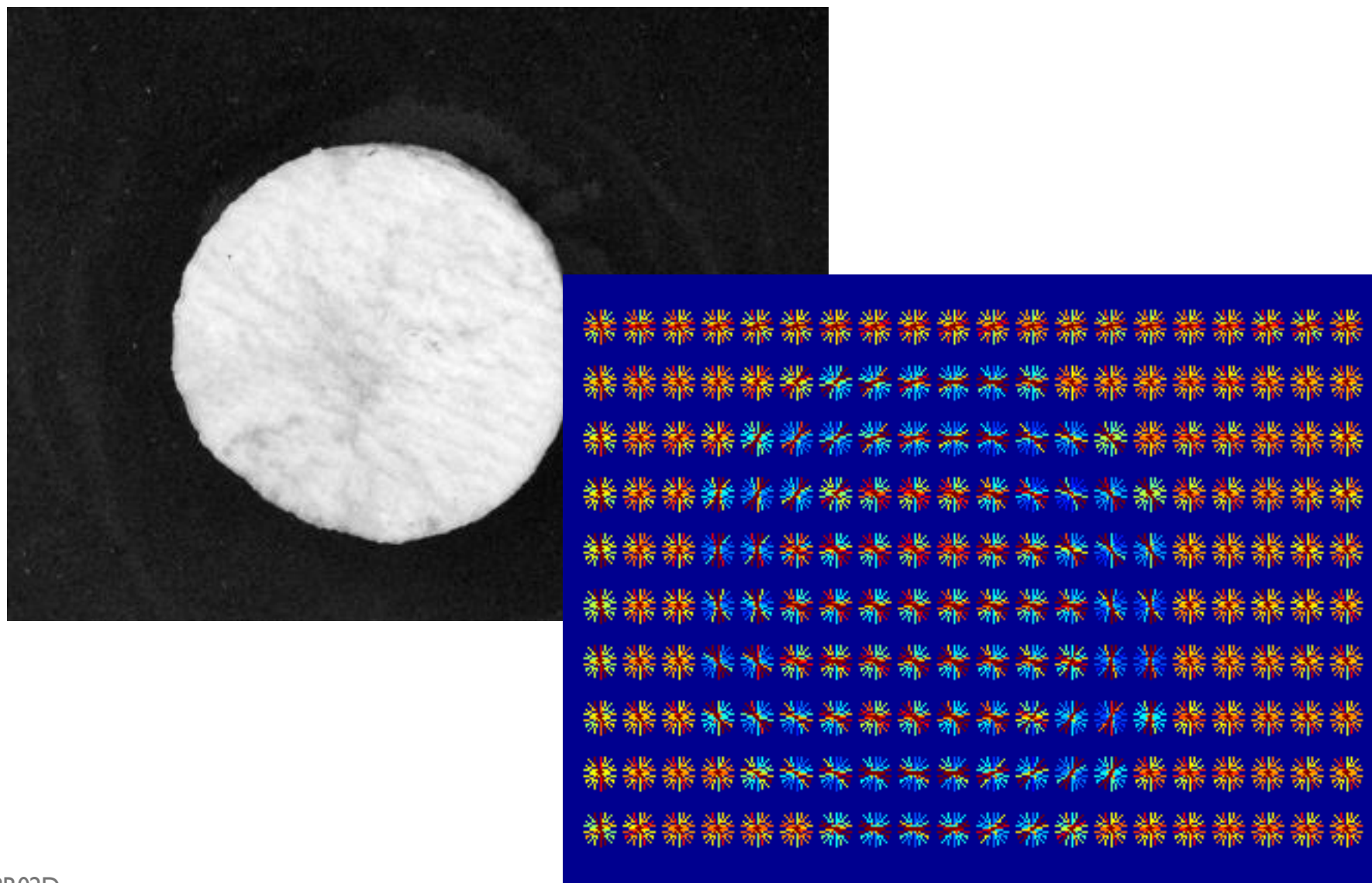
Descriptor de n elementos
 $n = \# \text{ celdas} \times \# \text{ bins}$





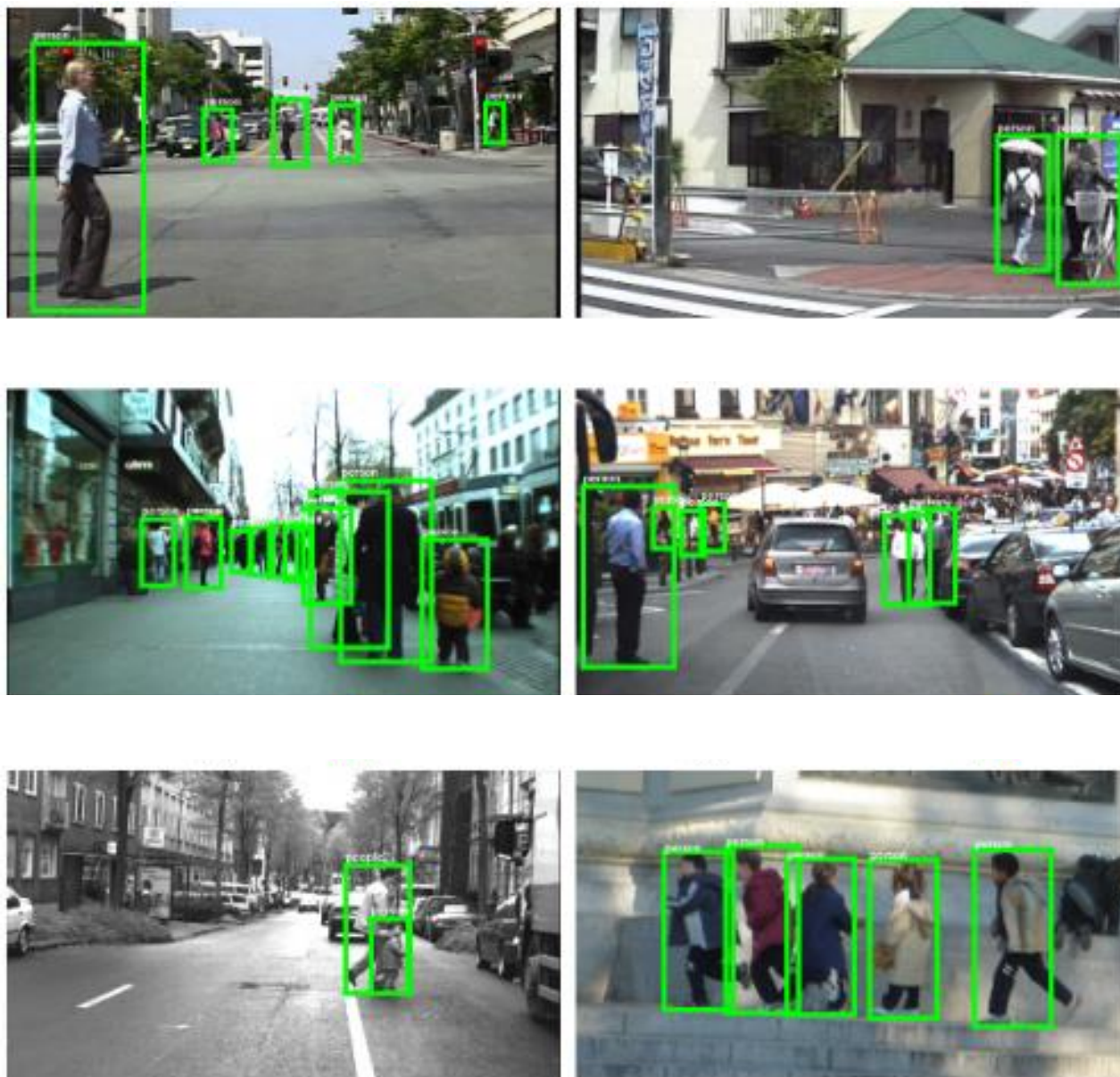


Ejemplo en foto de una papita



¿Cómo detectar peatones usando HoG?

La solución



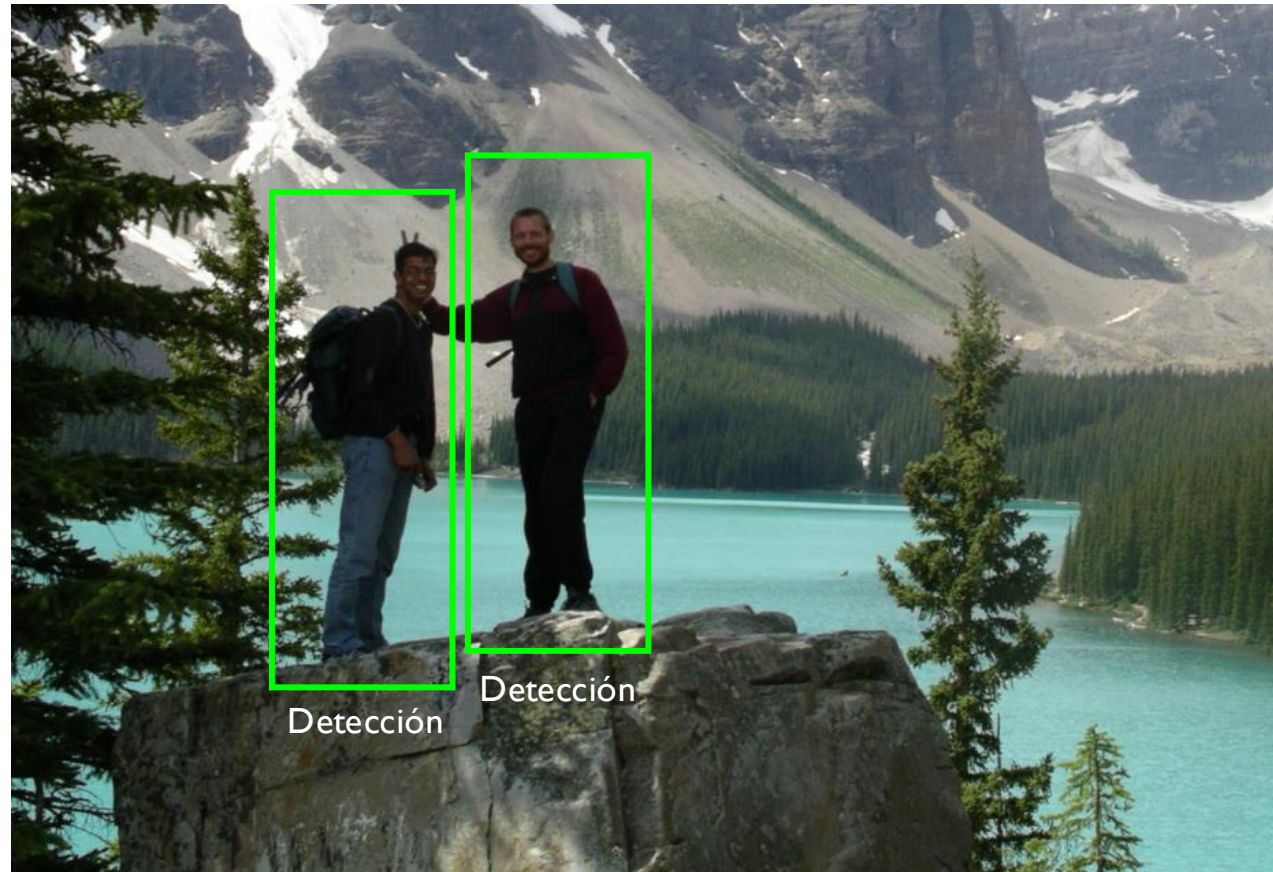
Estrategia: Sliding Windows



Estrategia: Sliding Windows



Estrategia: Sliding Windows



La idea clave

Diseñar un clasificador que pueda distinguir entre Peatones y No-peatones.

Clase positiva: Peatones



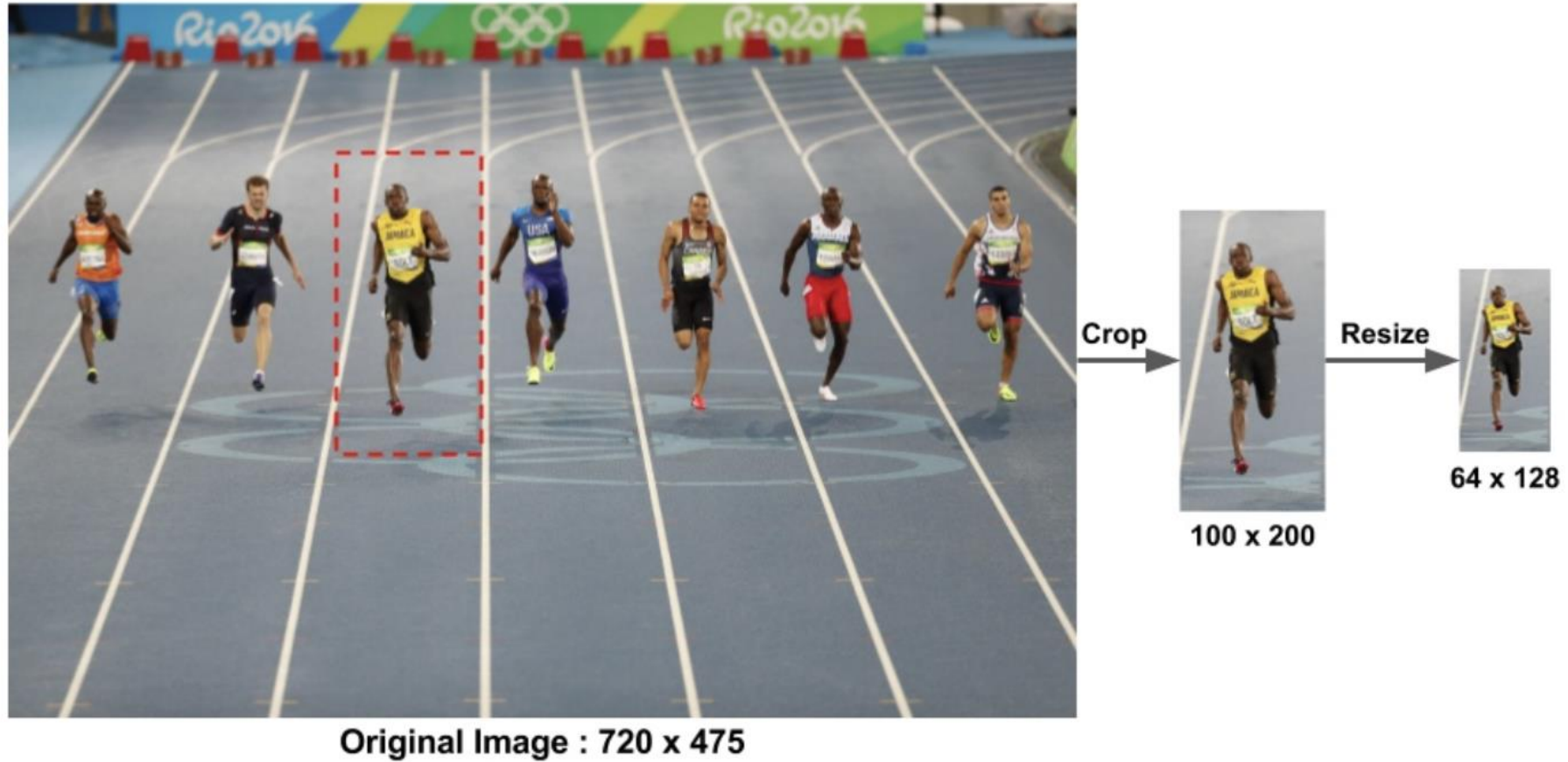
Peatón

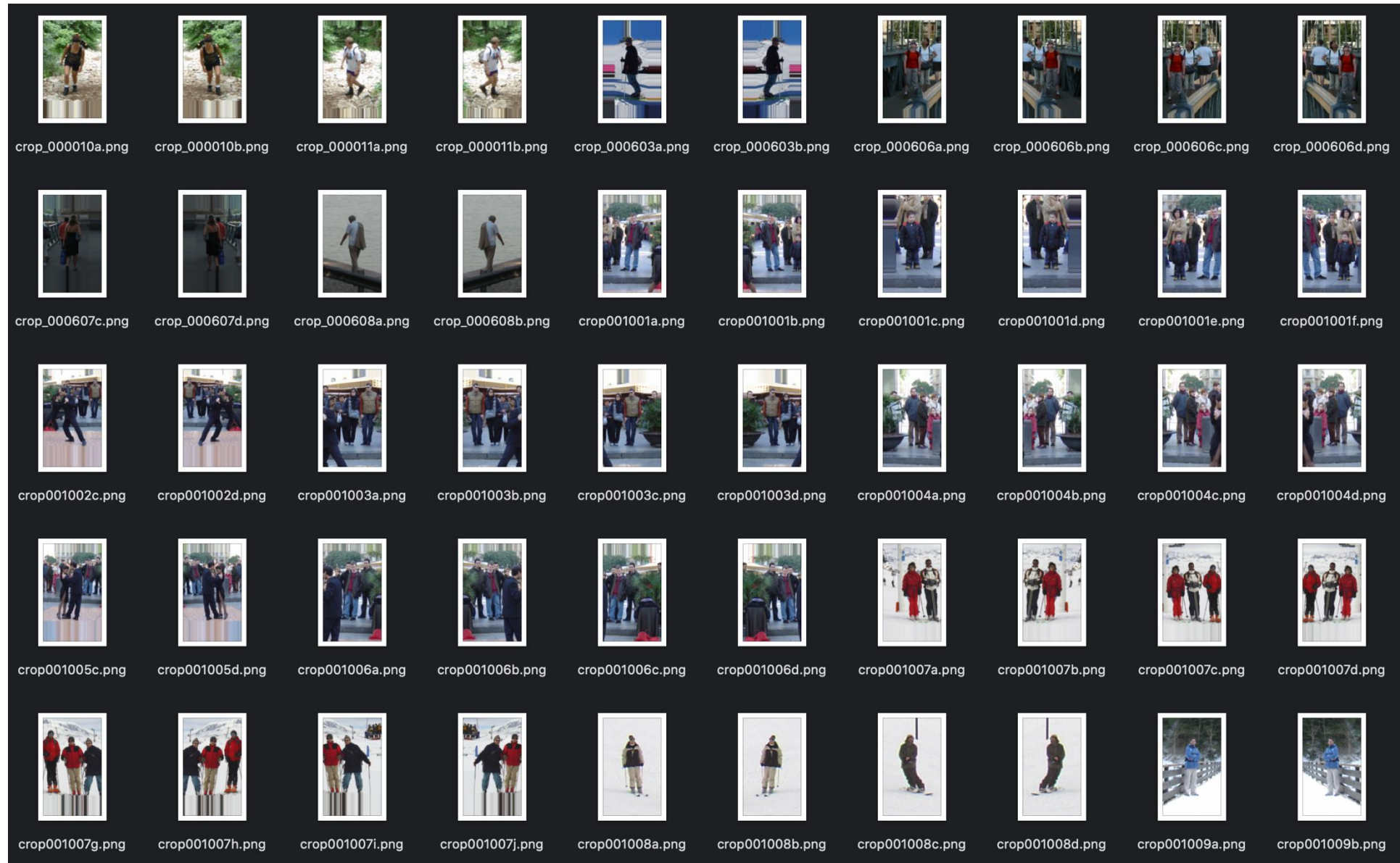
Clase negativa: No-peatones



No-peatones

Construcción del conjunto de datos: Clase positiva

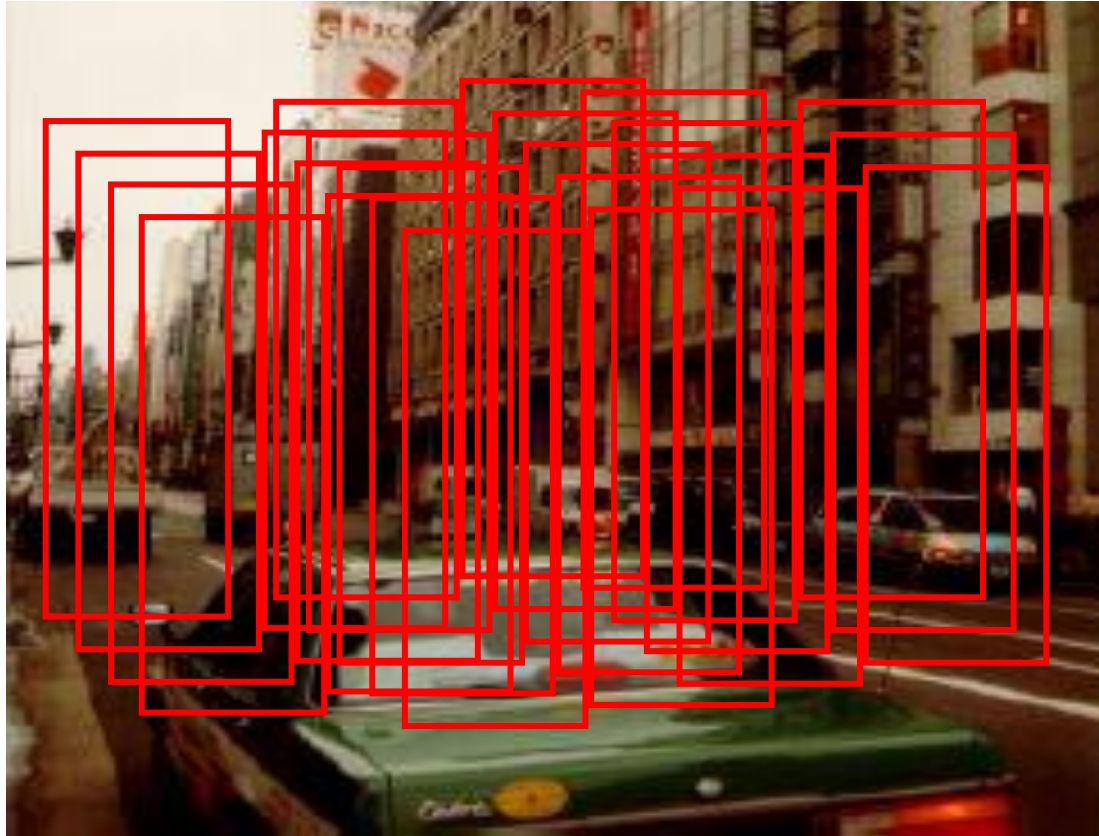




Construcción del conjunto de datos: Clase negativa



Construcción del conjunto de datos: Clase negativa



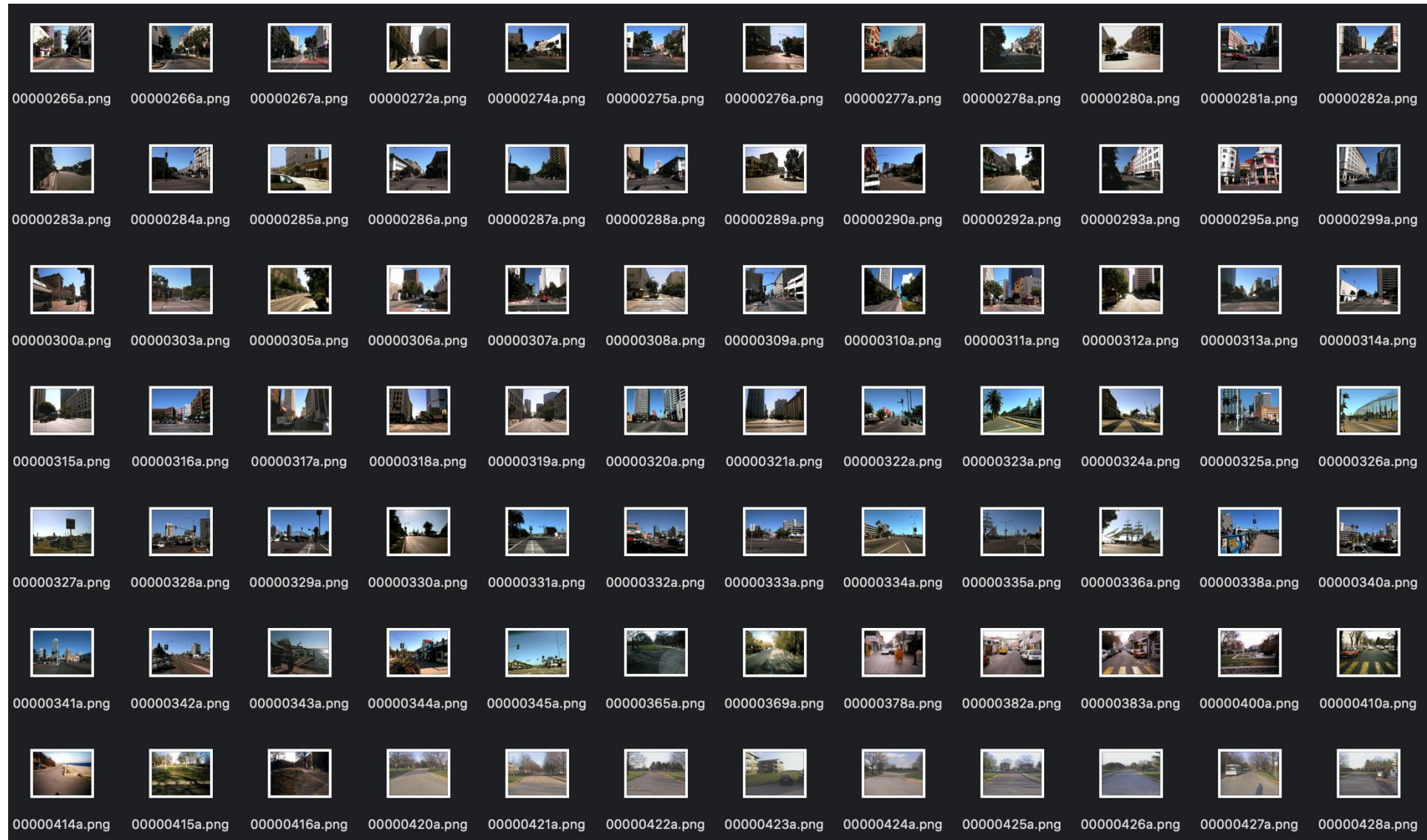
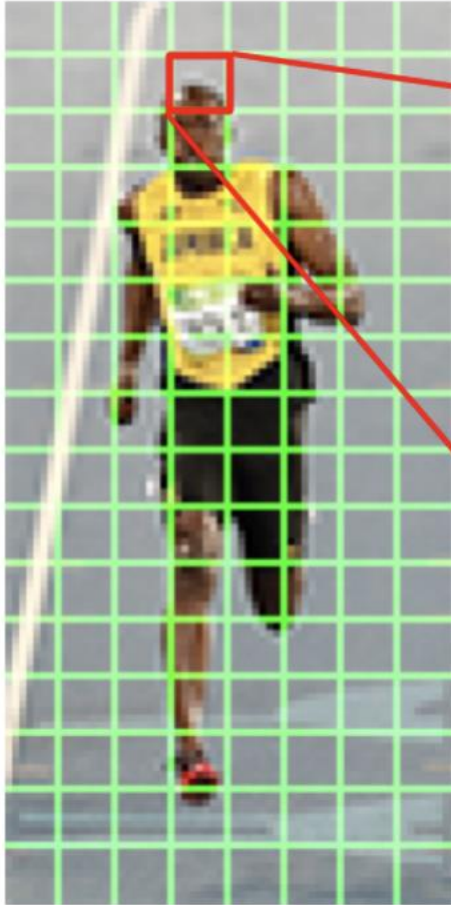
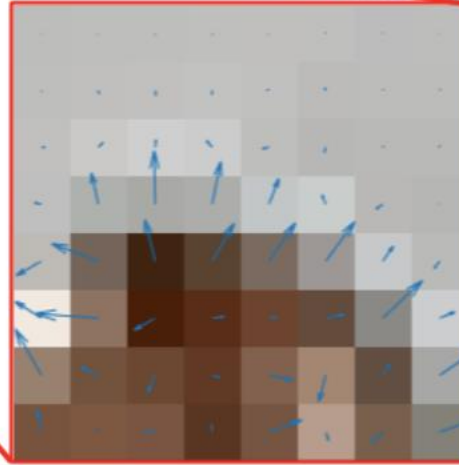


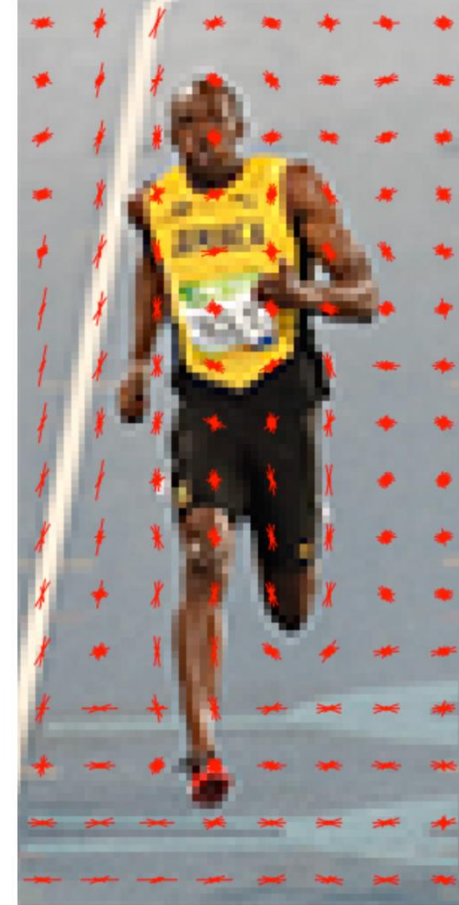
Imagen:
128 x 64 x 3 pixels



Extracción de
características



Descriptor
3780 elementos



Center : The RGB patch and gradients represented using arrows. Right : The gradients in the same patch represented as numbers

Dataset: Clase Positiva



Pedestrian	Element-1	Element-2	Element-3	Element-4	Element-5	...	Element-3780
1	0.3452	0.1151	0.2685	0.1342	0.2416		0.2301
2	0.2301	0.1534	0.2148	0.1611	0.2071		0.3068
3	0.1726	0.1918	0.1790	0.1879	0.1812		0.3836
4	0.1381	0.2301	0.1534	0.2148	0.1611		0.4603
5	0.1151	0.2685	0.1342	0.2416	0.1450		0.5370
6	0.0986	0.3068	0.1193	0.2685	0.1318		0.6137
7	0.0863	0.3452	0.1074	0.2953	0.1208		0.6904
8	0.0767	0.3836	0.0976	0.3222	0.1115		0.7671
9	0.0690	0.4219	0.0895	0.3490	0.1036		0.8438
10	0.0628	0.4603	0.0826	0.3759	0.0967		0.9205
11	0.0575	0.4986	0.0767	0.4027	0.0906		0.9972
12	0.0531	0.5370	0.0716	0.4296	0.0853		1.0740
13	0.0493	0.5753	0.0671	0.4564	0.0805		1.1507
14	0.0460	0.6137	0.0632	0.4833	0.0763		1.2274
15	0.0432	0.6520	0.0597	0.5101	0.0725		1.3041
16	0.0406	0.6904	0.0565	0.5370	0.0690		1.3808
17	0.0384	0.7288	0.0537	0.5638	0.0659		1.4575
18	0.0363	0.7671	0.0511	0.5907	0.0630		1.5342
19	0.0345	0.8055	0.0488	0.6175	0.0604		1.6109
20	0.0329	0.8438	0.0467	0.6444	0.0580		1.6876
21	0.0314	0.8822	0.0447	0.6712	0.4851		1.7644
22	0.0300	0.9205	0.0430	0.0802	0.6444		1.8411
23	0.0288	0.9589	0.3593	0.0604	0.4660		1.9178
24	0.0276	0.1146	0.4773	0.0835	0.3625		0.2293
25	0.2310	0.0863	0.3452	0.1074	0.2953		0.1726
:	:	:	:	:	:		:
:	:	:	:	:	:		:
100000	0.1230	0.0493	0.5753	0.0671	0.4564		0.0986

Dataset: Clase Negativa



No-Pedestrian	Element-1	Element-2	Element-3	Element-4	Element-5	...	Element-3780
1	0.7452	0.2484	0.5796	0.2898	0.5216		0.4968
2	0.4968	0.3312	0.4637	0.3478	0.4471		0.6624
3	0.3726	0.4140	0.3864	0.4057	0.3912		0.8280
4	0.2981	0.4968	0.3312	0.4637	0.3478		0.9936
5	0.2484	0.5796	0.2898	0.5216	0.3130		1.1592
6	0.2129	0.6624	0.2576	0.5796	0.2845		1.3248
7	0.1863	0.7452	0.2318	0.6376	0.2608		1.4904
8	0.1656	0.8280	0.2108	0.6955	0.2408		1.6560
9	0.1490	0.9108	0.1932	0.7535	0.2236		1.8216
10	0.1355	0.9936	0.1783	0.8114	0.2087		1.9872
11	0.1242	1.0764	0.1656	0.8694	0.1956		2.1528
12	0.1146	1.1592	0.1546	0.9274	0.1841		2.3184
13	0.1065	1.2420	0.1449	0.9853	0.1739		2.4840
14	0.0994	1.3248	0.1364	1.0433	0.1647		2.6496
15	0.0932	1.4076	0.1288	1.1012	0.1565		2.8152
16	0.0877	1.4904	0.1220	1.1592	0.1490		2.9808
17	0.0828	1.5732	0.1159	1.2172	0.1423		3.1464
18	0.0784	1.6560	0.1104	1.2751	0.1361		3.3120
19	0.0745	1.7388	0.1054	1.3331	0.1304		3.4776
20	0.0710	1.8216	0.1008	1.3910	0.1252		3.6432
21	0.0677	1.9044	0.0966	1.4490	0.4851		3.8088
22	0.0648	1.9872	0.0927	0.3740	1.3910		3.9744
23	0.0621	2.0700	0.3593	0.1304	1.0060		4.1400
24	0.0596	0.5342	1.0304	0.1803	0.7825		1.0684
25	0.2310	0.1863	0.7452	0.2318	0.6376		0.3726
:	:	:	:	:	:		:
:	:	:	:	:	:		:
800000	0.9230	0.1065	1.2420	0.1449	0.9853		0.2129

Training

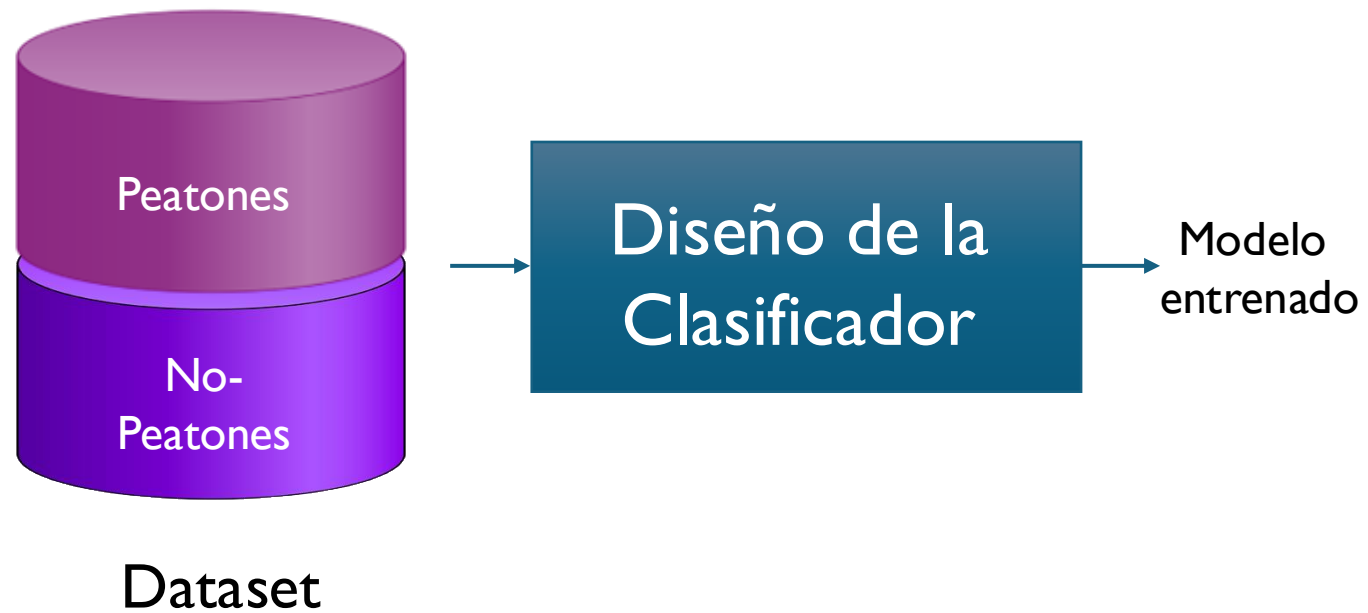
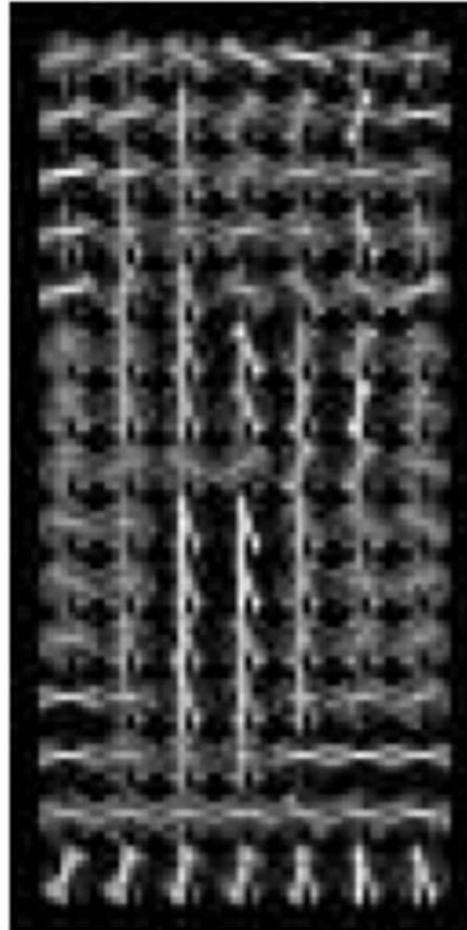


Imagen:
128 x 64 x 3 pixels



Descriptor
3780 elementos



Output:
0 / 1

Pedestrian

Extracción de Características
Usando HoG

Clasificación
Usando Modelo Entrenado



Ejemplo

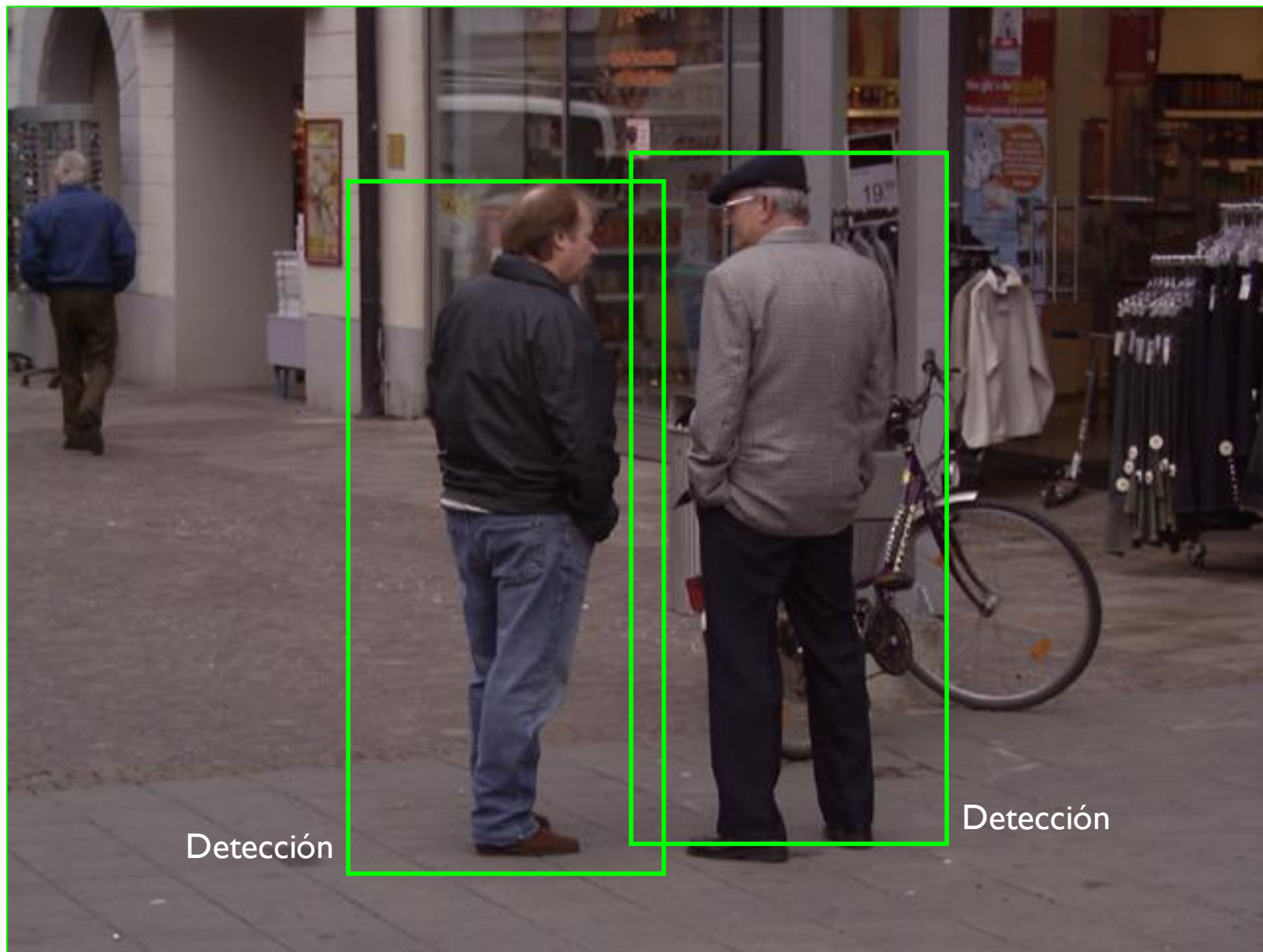


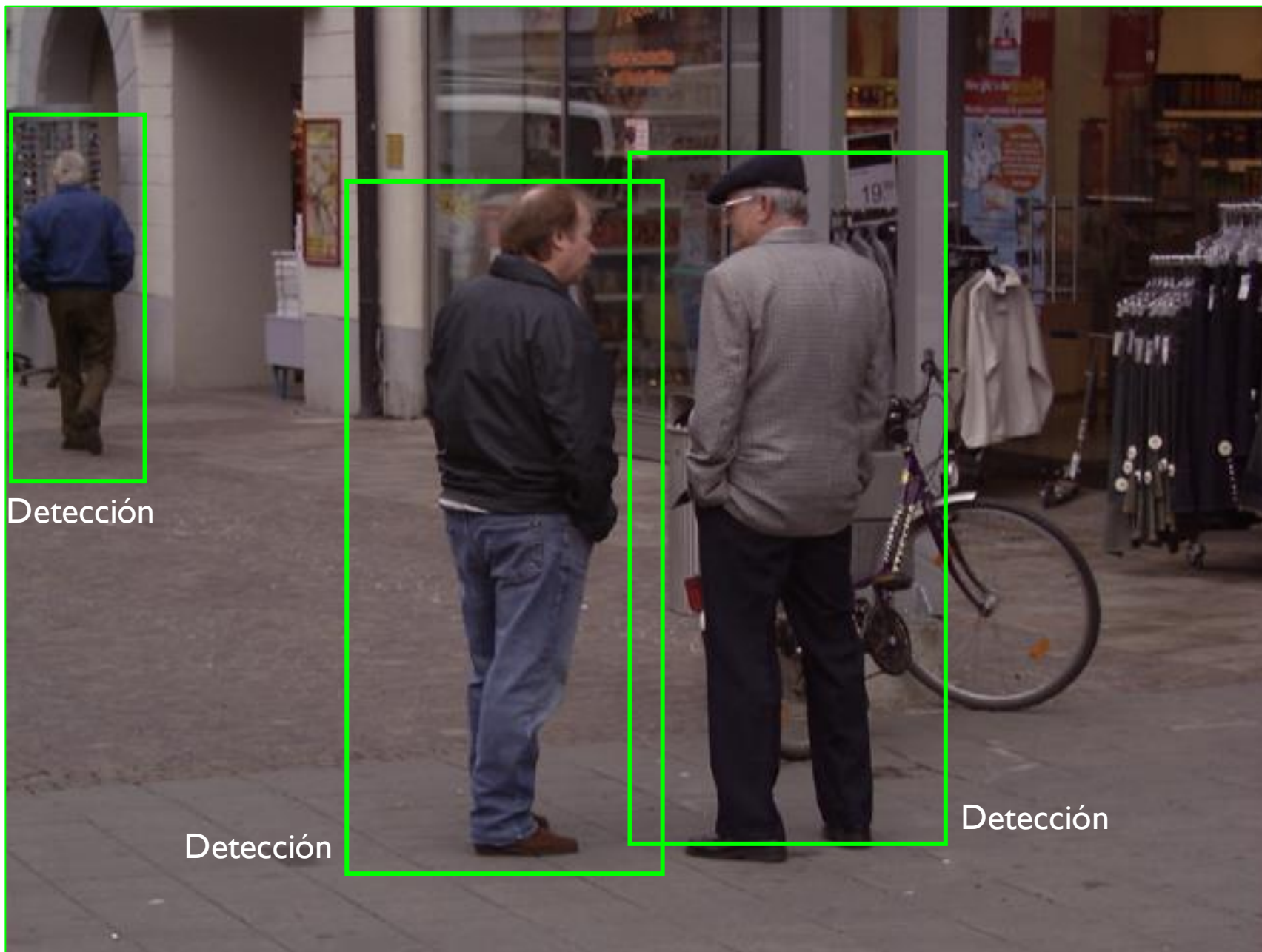


Detección

Ejemplo







Detección

Detección

Detección

Características de Intensidad

Local Binary Patterns (LBP)

Ojala, T., Pietikainen, M., 2 Maenpaa, T. (2002). [Multiresolution gray-scale and rotation invariant texture classification](#)

(c) D.Mery - PR02D [with local binary patterns](#). IEEE Transactions on pattern analysis and machine intelligence, 24(7), 971-987.

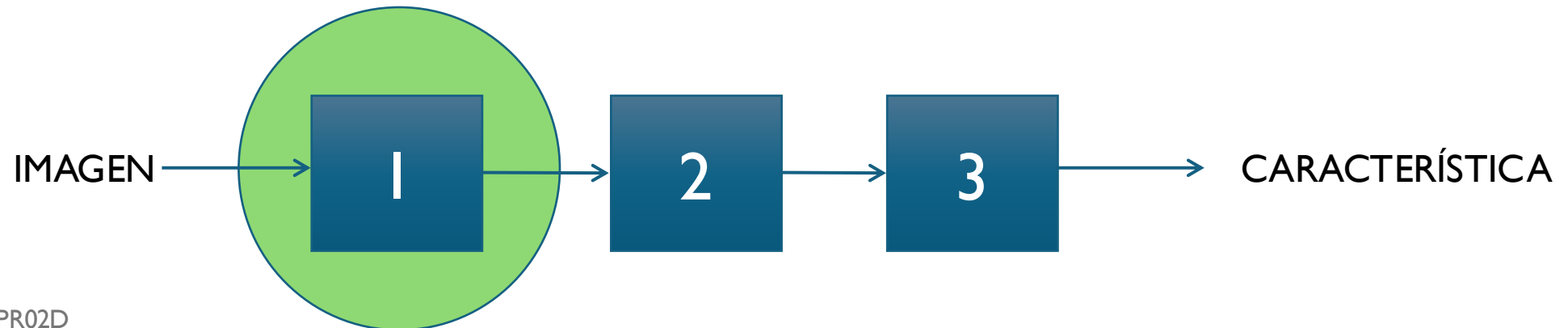
Local Binary Patterns

1. Codificación
2. Mapeo
3. Histograma



Local Binary Patterns

1. Codificación
2. Mapeo
3. Histograma





4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

<		

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0		

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	$\geq$	

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
		0

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
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		0

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
		0
	1	0

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
		0
1	1	0

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
1		0
1	1	0

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
1		0
1	1	0

x

1	2	4
128	+	8
64	32	16

$$= 2+4+32+64+128 = 230$$

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

4	6	9
9	6	4
9	6	2

0	1	1
1		0
1	1	0

x

1	2	4
128	+	8
64	32	16

$$= 2+4+32+64+128 = 230$$

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230				

4	6	9
9	6	4
9	6	2

0	1	1
1		0
1	1	0

x

1	2	4
128	+	8
64	32	16

$$= 2+4+32+64+128 = 230$$

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	?			

6	9	6
6	4	9
6	2	2

x

1	2	4
128	+	8
64	32	16

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207			

6	9	6
6	4	9
6	2	2

1	1	1
1		1
1	0	0

x

1	2	4
128	+	8
64	32	16

$$= 1 + 2 + 4 + 8 + 64 + 128 = 207$$

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	?		

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25		

9	6	4
4	9	9
2	2	9

1	0	0
0		1
0	0	1

x

1	2	4
128	+	8
64	32	16

$$= 1 + 8 + 16 = 25$$

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25	168	

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25	168	
	243				

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25	168	
	243	255			

4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25	168	
	243	255	255		

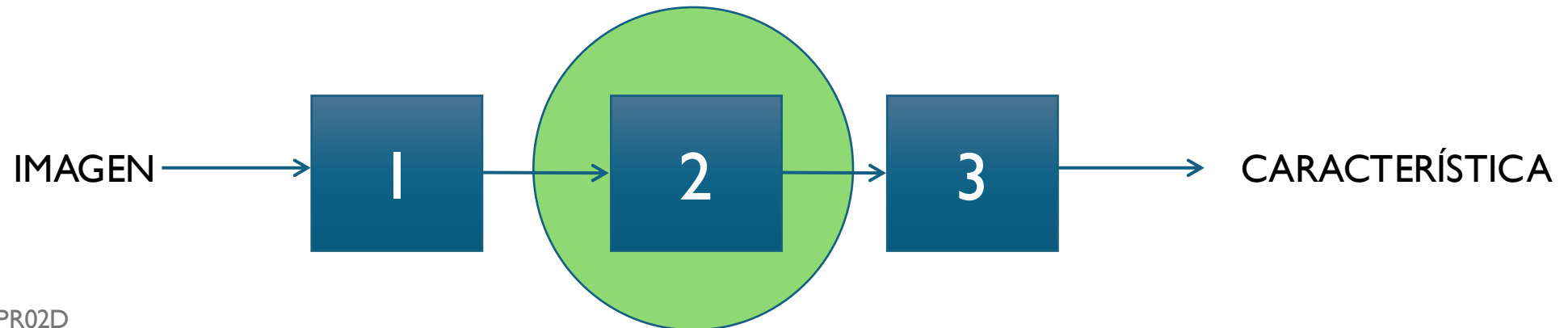
4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10



	230	207	25	168	
	243	255	255	119	

Local Binary Patterns

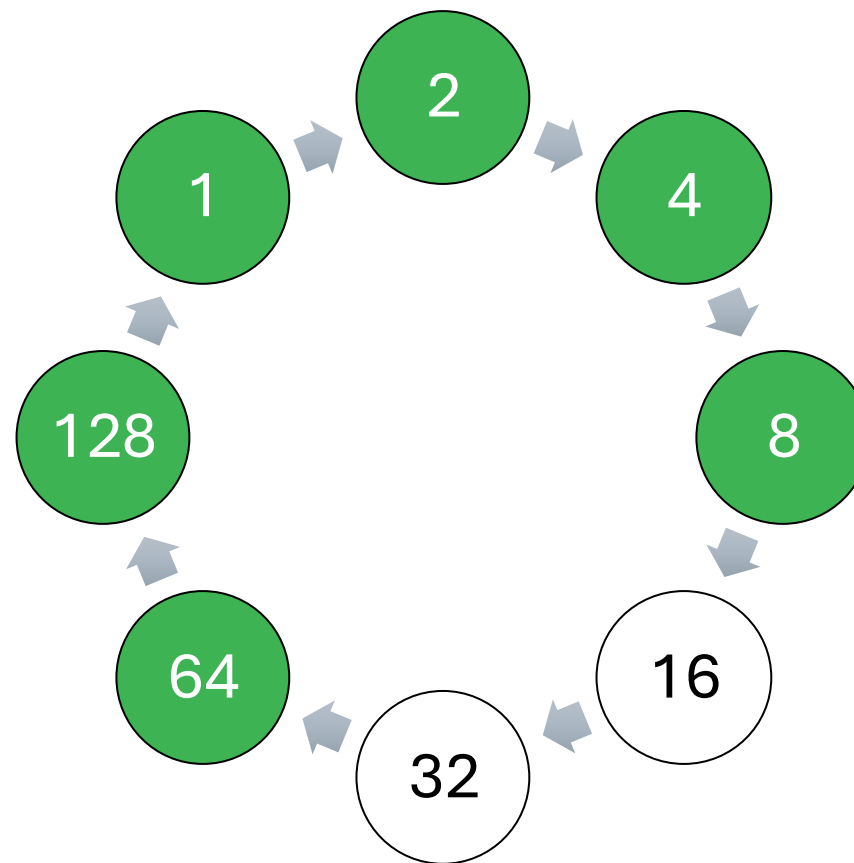
1. Codificación
- 2. Mapeo**
3. Histograma



6	9	6
6	4	9
6	2	2



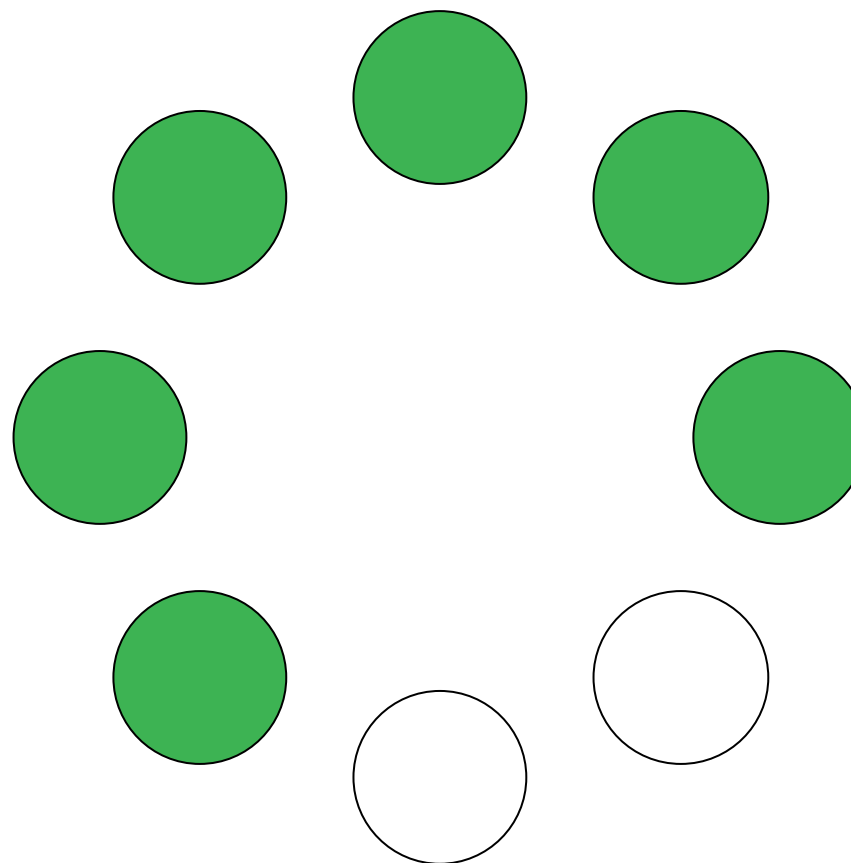
1	1	1
1		1
1	0	0



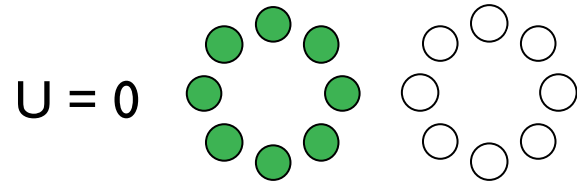
6	9	6
6	4	9
6	2	2



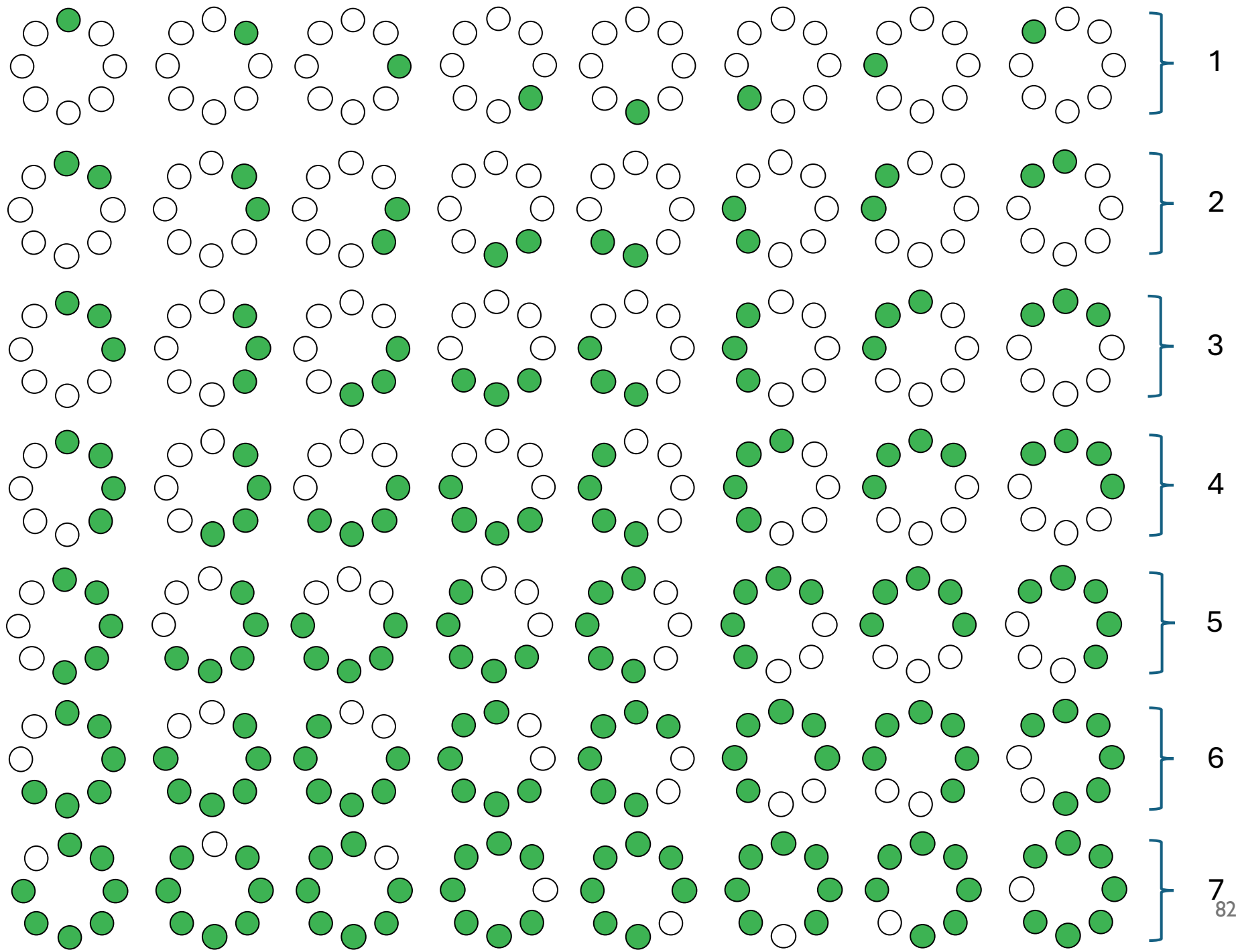
1	1	1
1		1
1	0	0



Patrones uniformes



$U = 2$



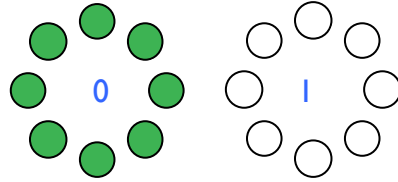
Patrones uniformes

$2 + 56 = 58$
patrones

$U = 0$

2
patrones

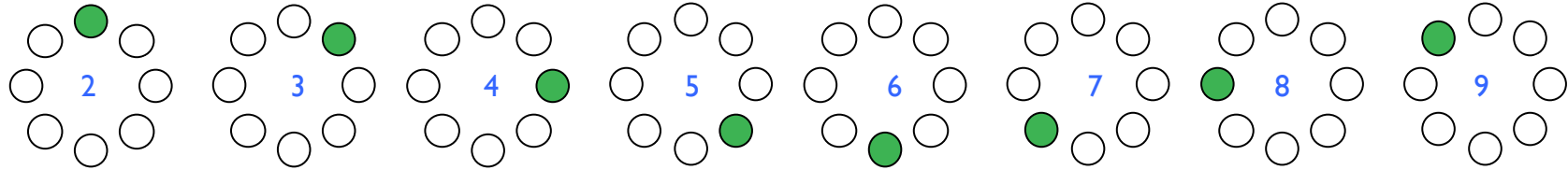
$\{0, 1\}$



$U = 2$

$8 \times 7 = 56$
patrones

$\{2, 3, \dots, 57\}$

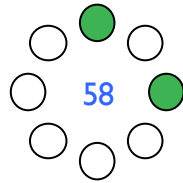


Patrones no uniformes

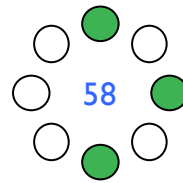
$256 - 58 = 198$
patrones

$\{58\}$

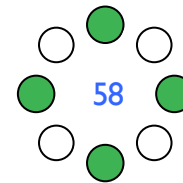
$U = 4$



$U = 6$



$U = 8$



4	6	9	6	4	6
9	6	4	9	9	9
9	6	2	2	9	2
10	10	10	10	10	10

IMAGEN



	230	207	25	168	
	243	255	255	119	

IMAGEN
CODIFICADA

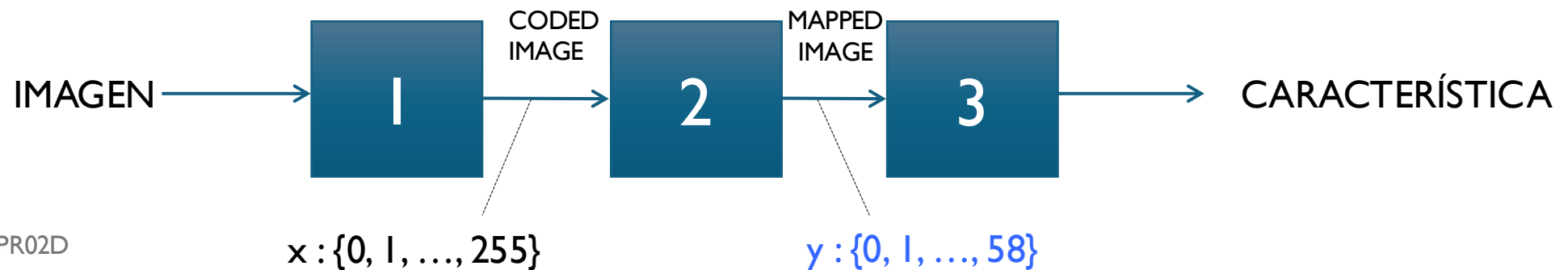


	58	46	58	58	
	23	1	1	58	

IMAGEN
MAPEADA

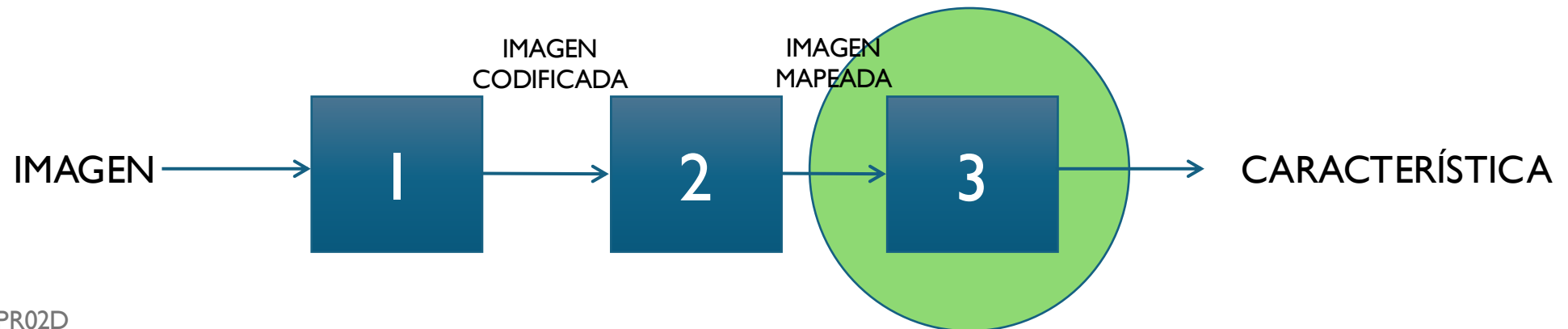
Local Binary Patterns

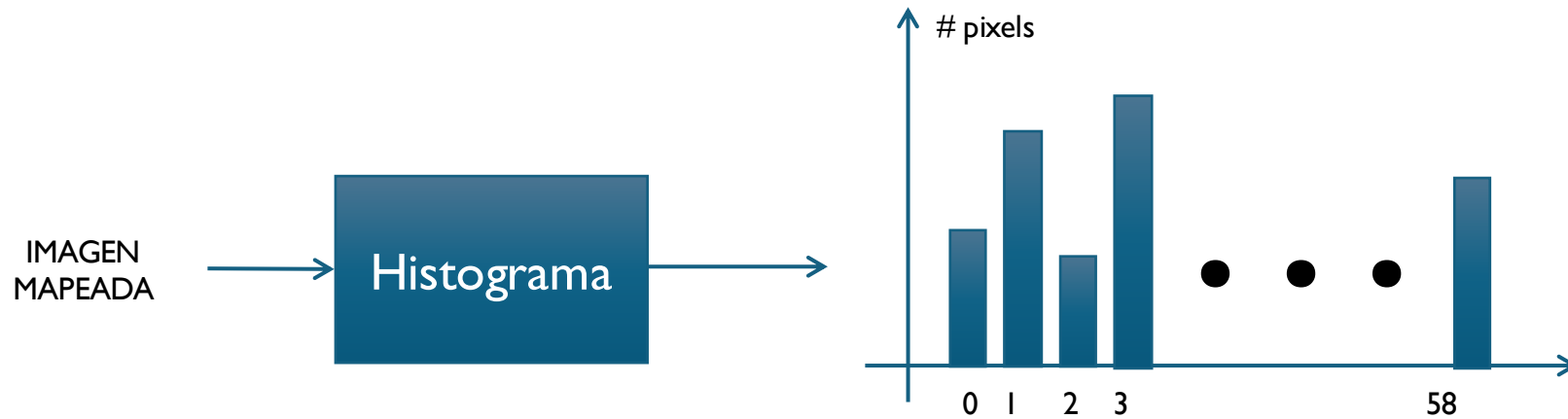
1. Codificación
- 2. Mapeo**
3. Histograma



Local Binary Patterns

1. Codificación
2. Mapeo
- 3. Histograma**

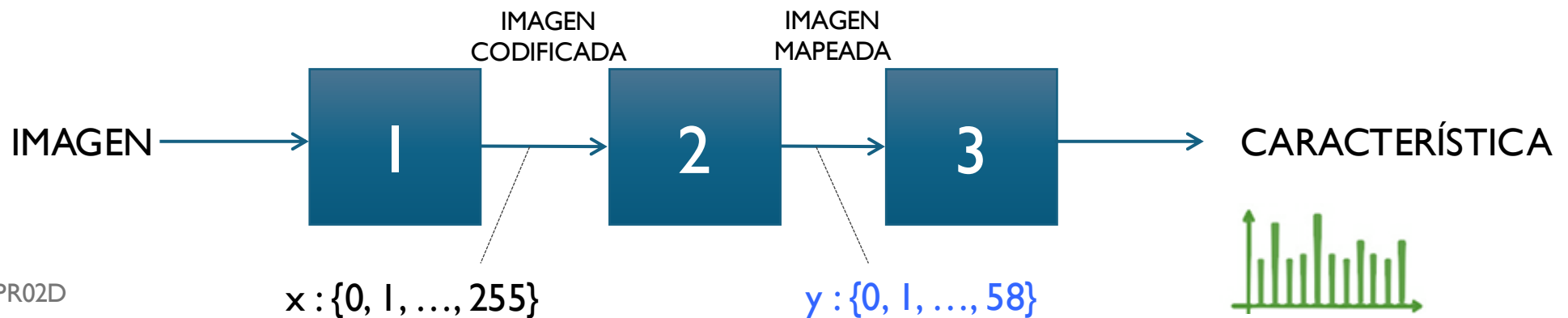




La imagen se describe como un vector de 59 elementos. Imágenes similares tienen características similares de LBP!!

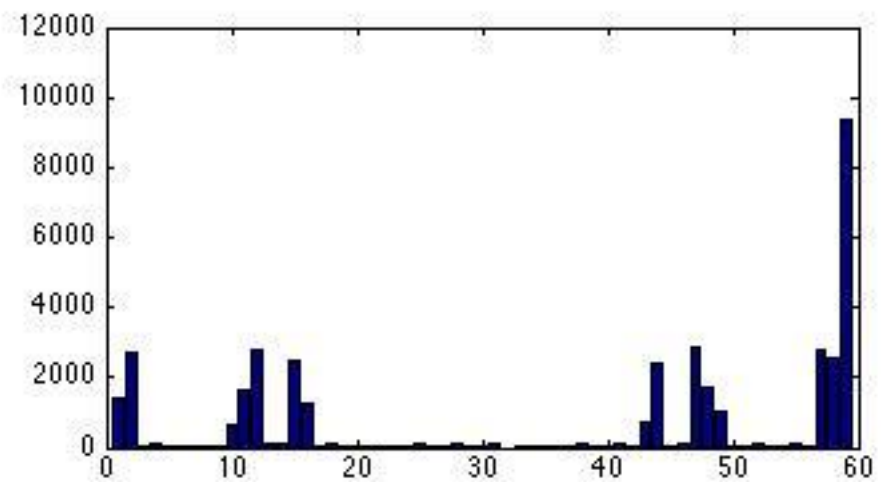
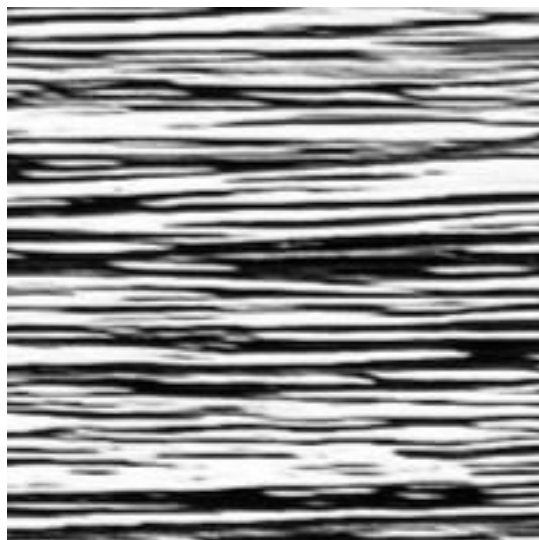
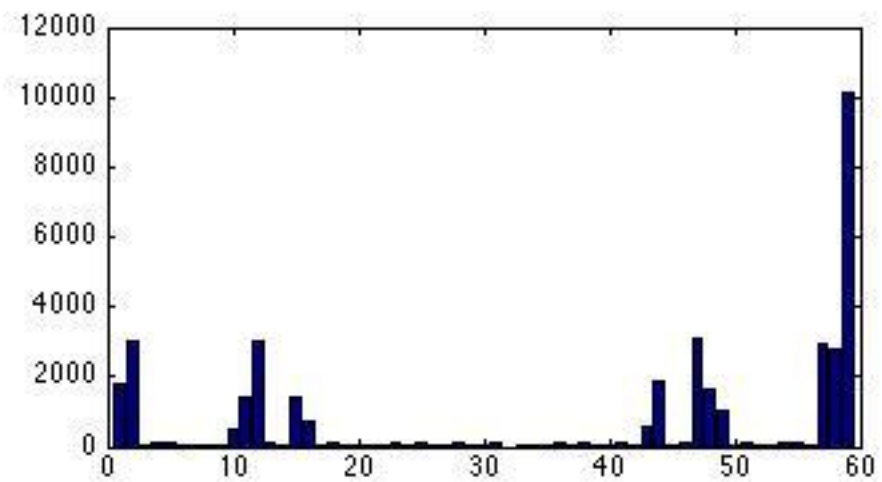
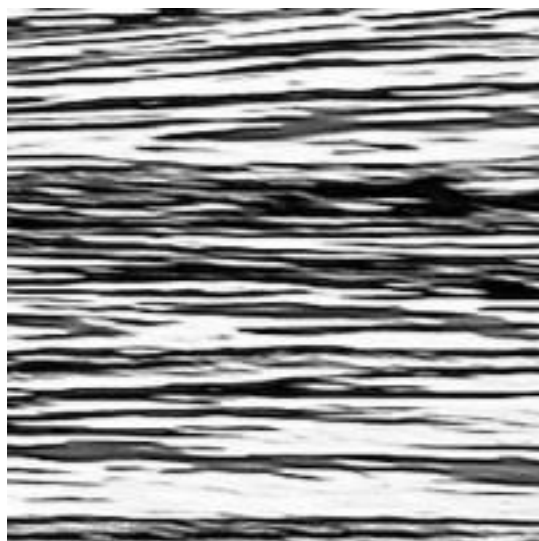
Local Binary Patterns

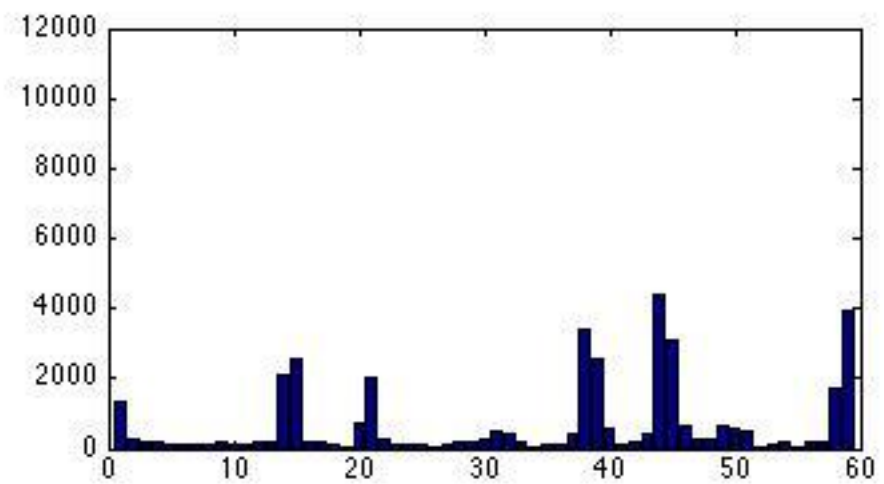
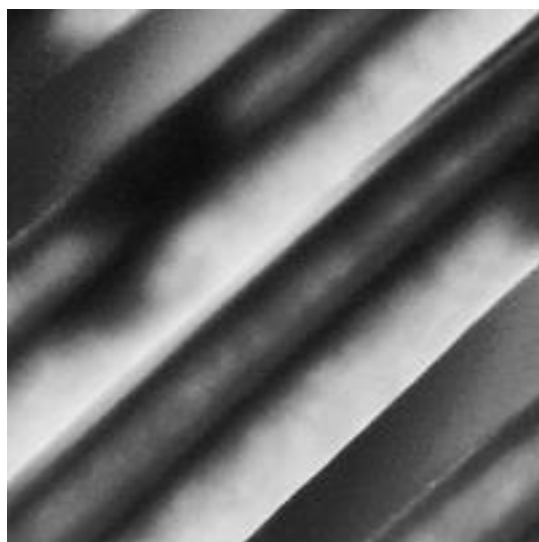
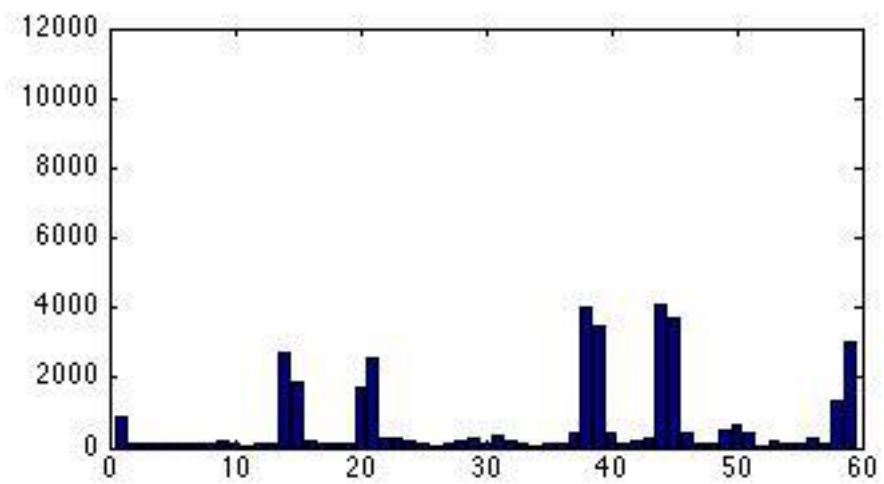
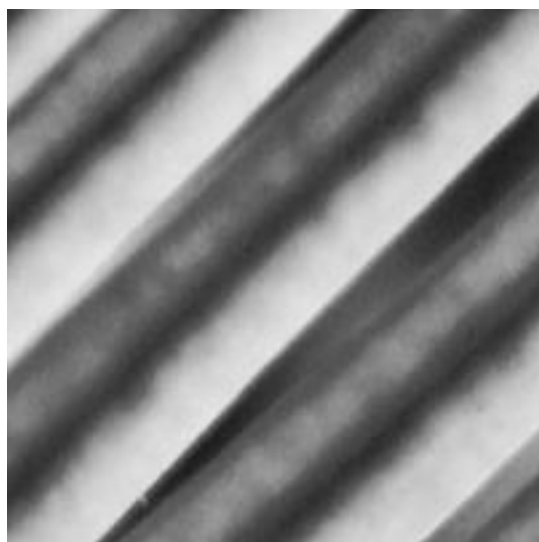
1. Codificación
2. Mapeo
3. Histograma

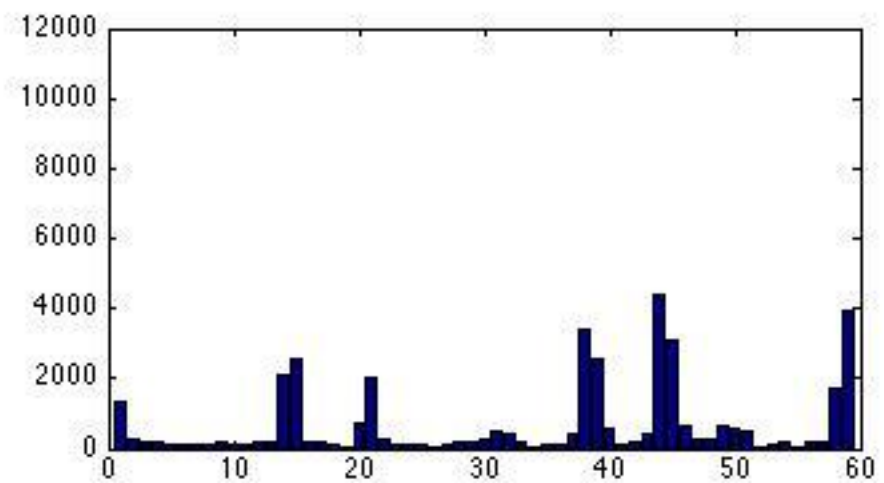
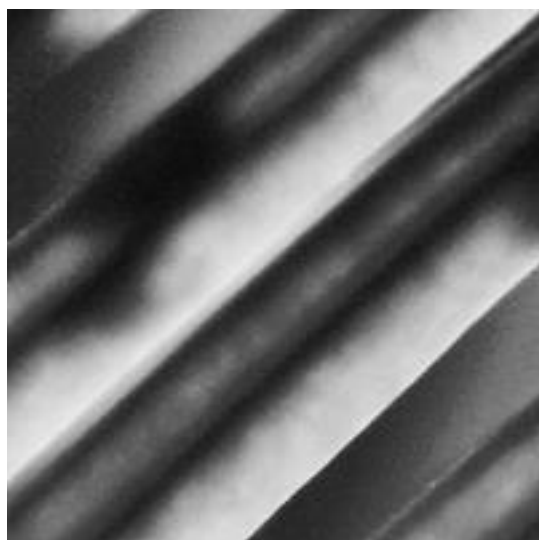
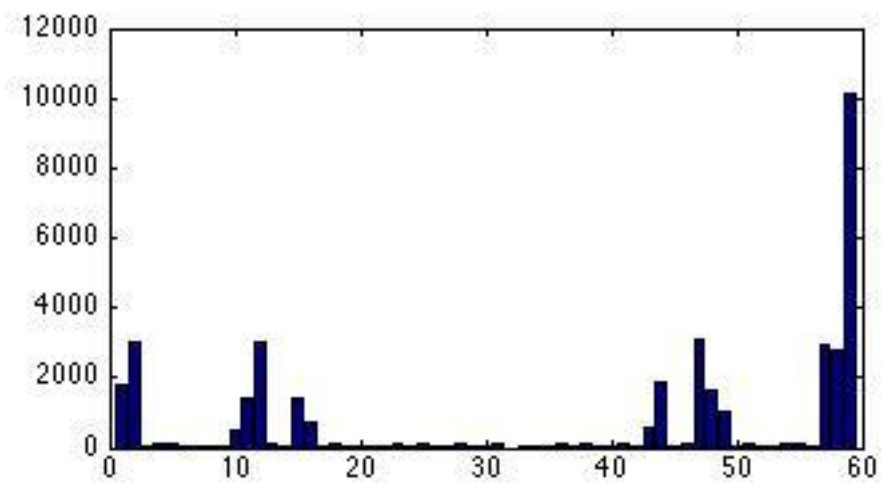
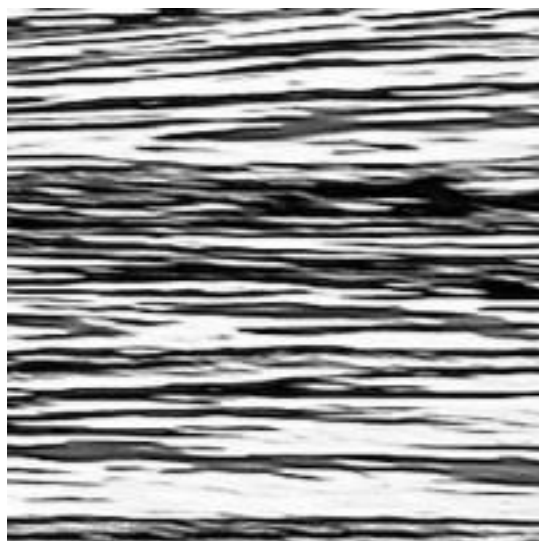


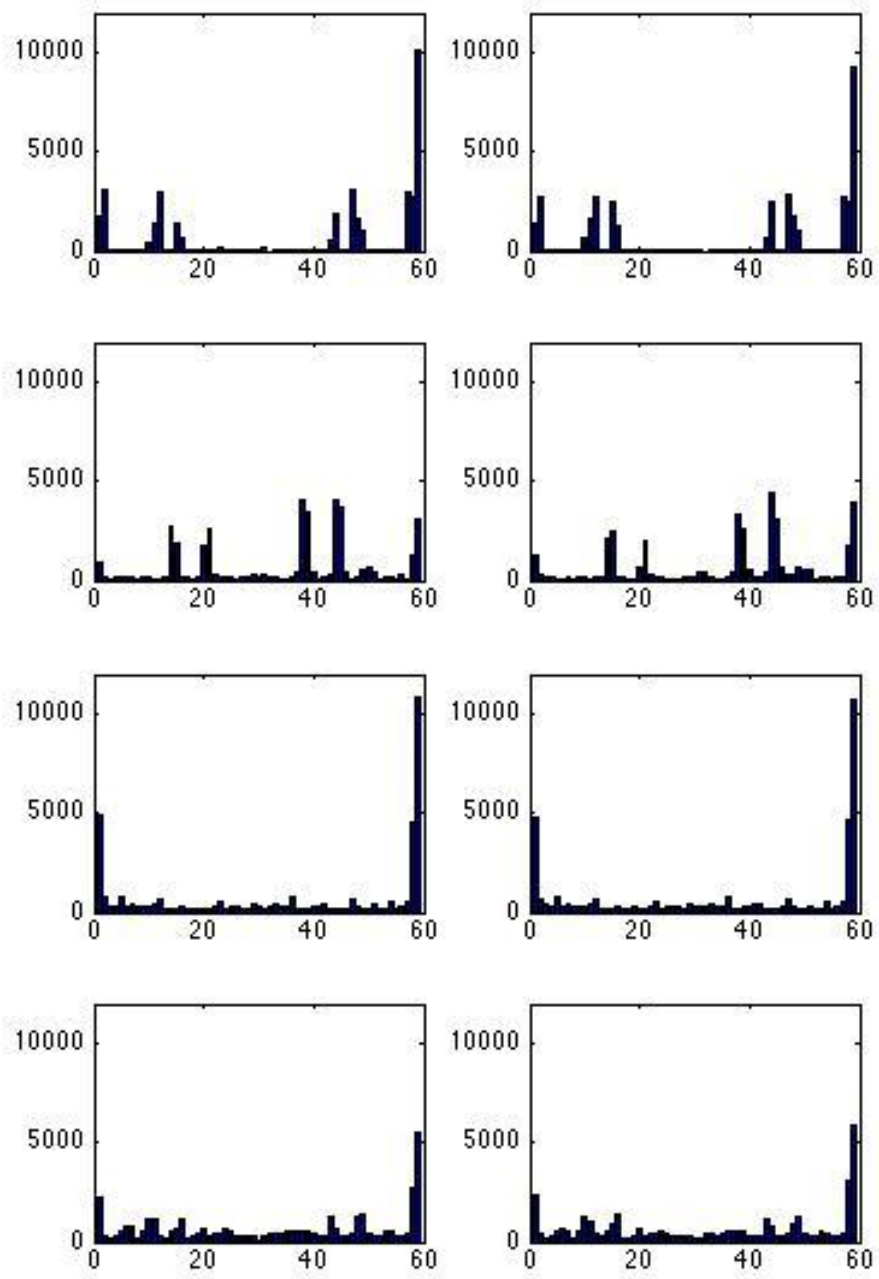
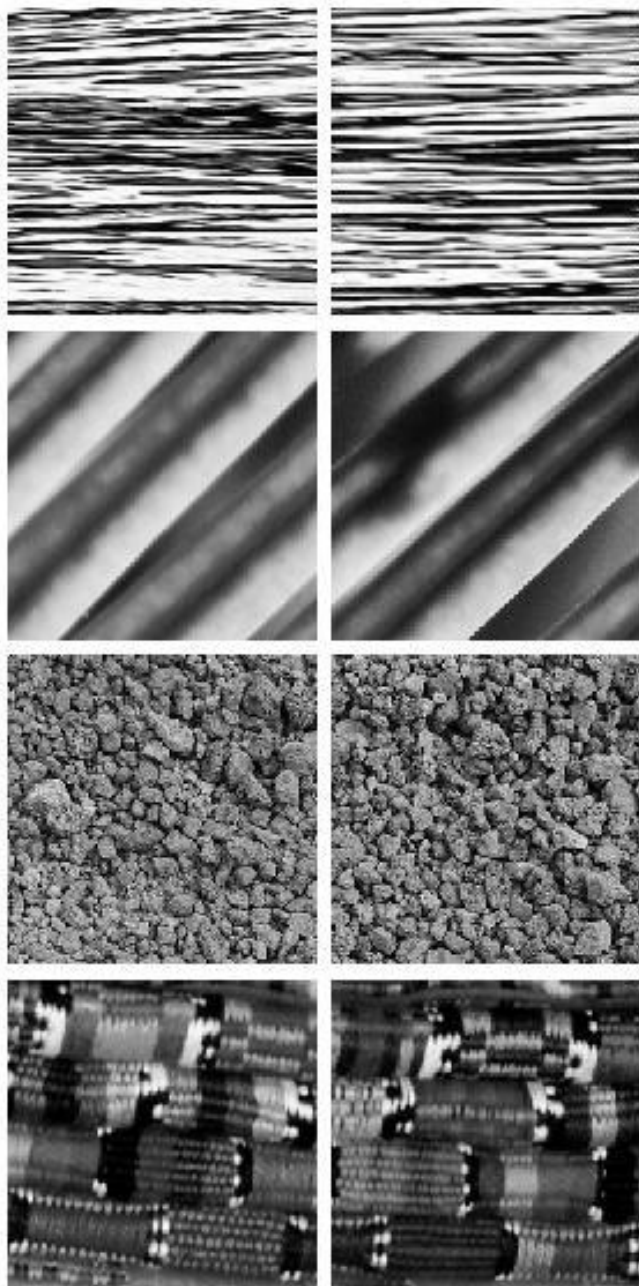
Local Binary Patterns

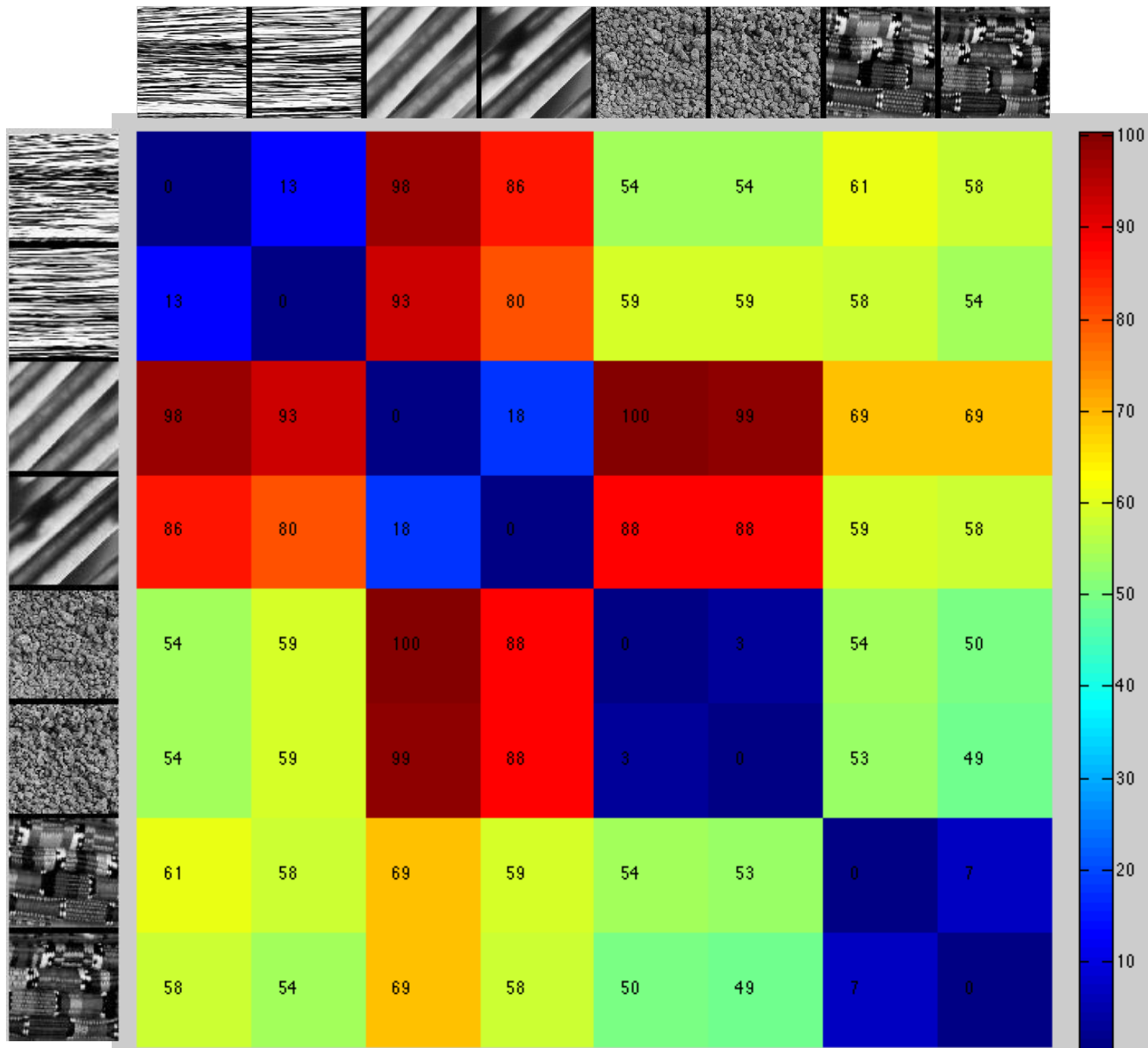
Ejemplos con
Texturas





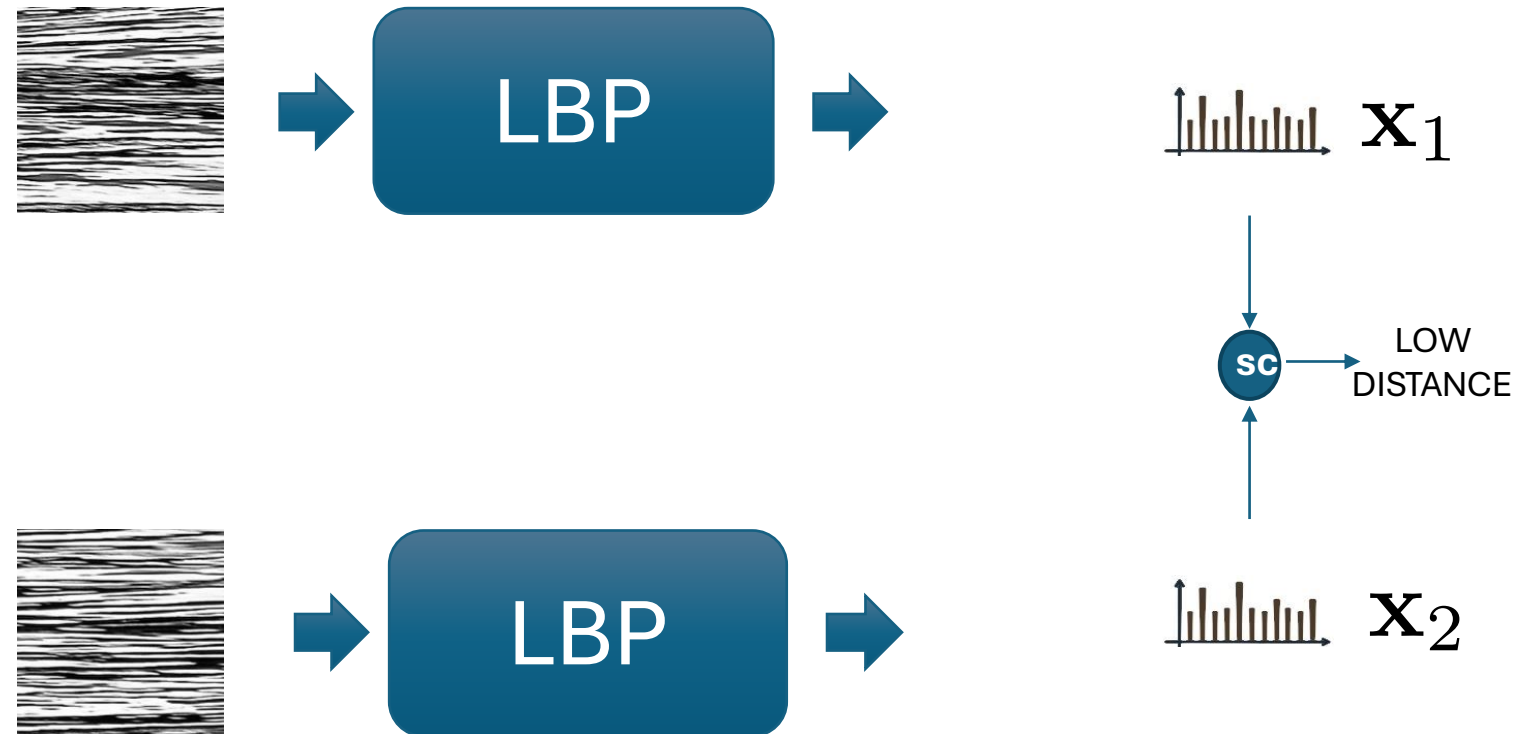






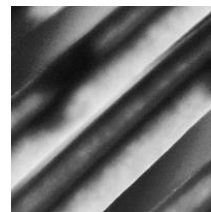
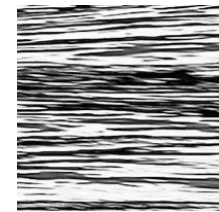
LBP en la práctica

FOTOS DE LA MISMA TEXTURA



LBP en la práctica

FOTOS DE TEXTURAS DISTINTAS



DISTANCIA
EUCLIDEANA:
mide qué tan distintos
son los descriptores



SC

HIGH
DISTANCE

Local Binary Patterns

Ejemplo en
Reconocimiento Facial



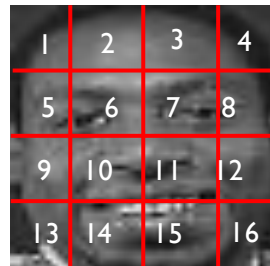
En el conjunto de entrenamiento hay C clases (C personas).

Para cada clase tenemos n imágenes de entrenamiento.

En este ejemplo hay 40 clases con 9 imágenes cada una.

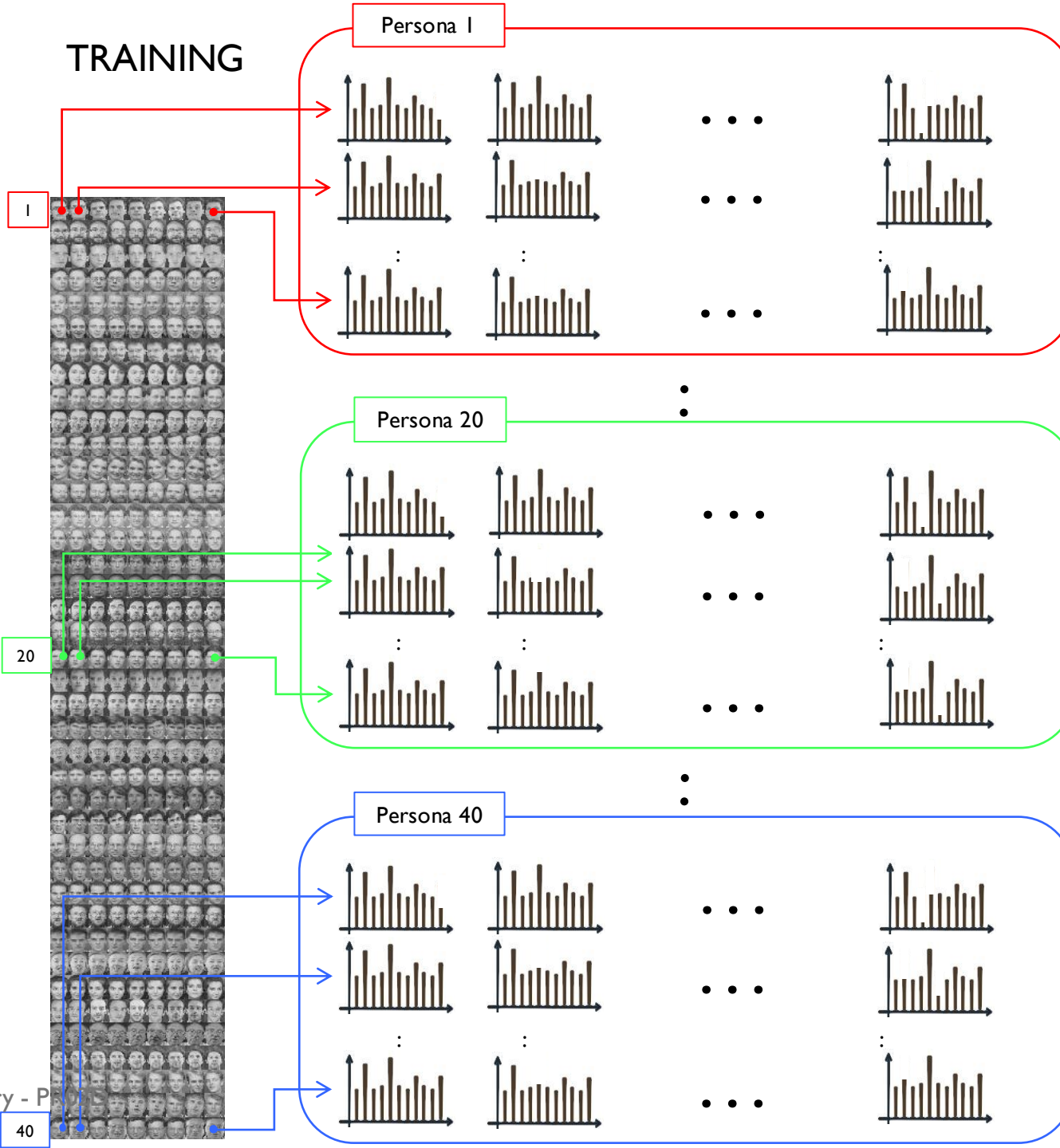
Cada imagen que usamos con $w \times w$ particiones

En cada partición extraemos la característica LBP



Una cara se describe usando una característica de $16 \times 59 = 944$ elementos

TRAINING

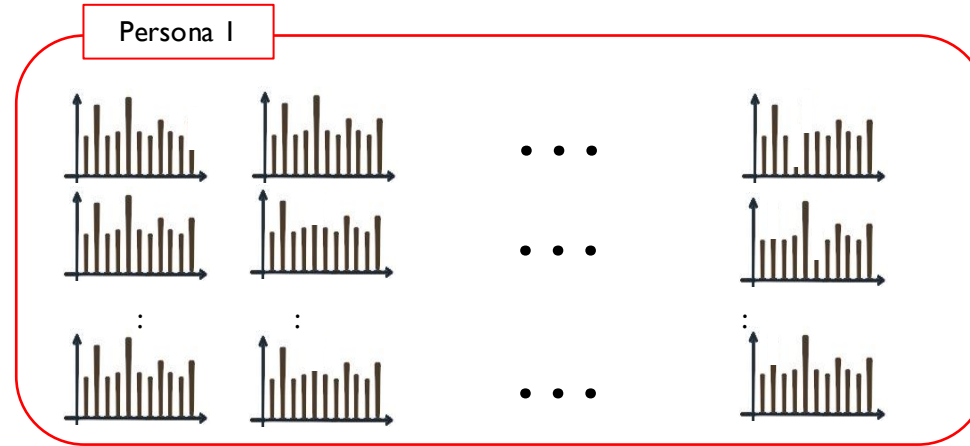


Datos de entrenamiento:

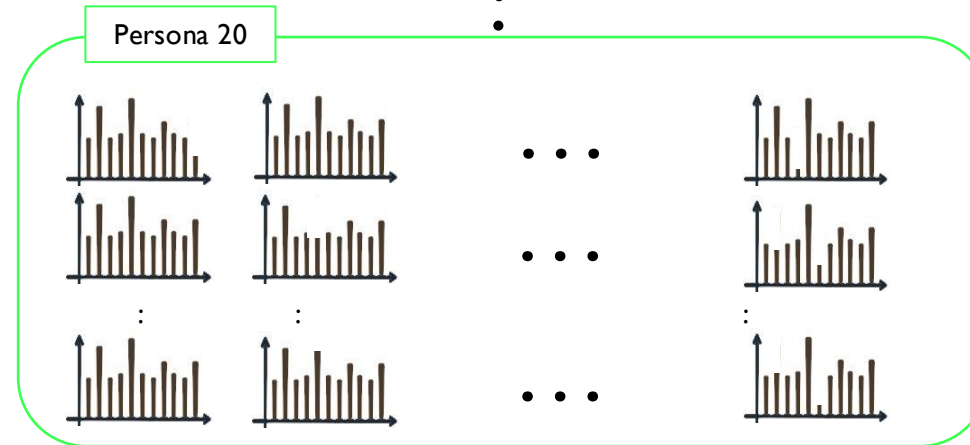
Tabla con:

$9 \times 40 = 360$ filas

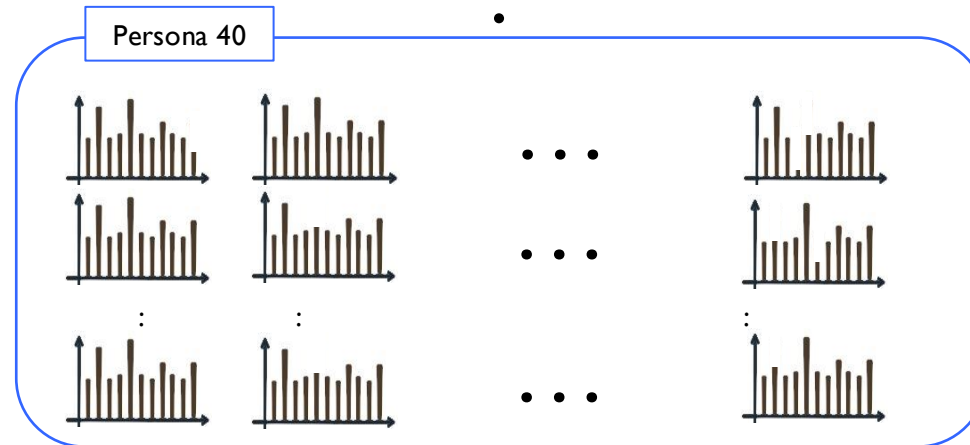
$16 \times 59 = 944$ columnas



•
•



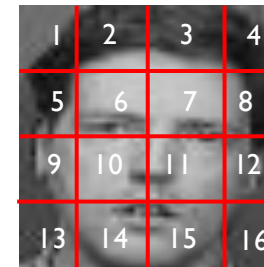
•
•



Persona 1

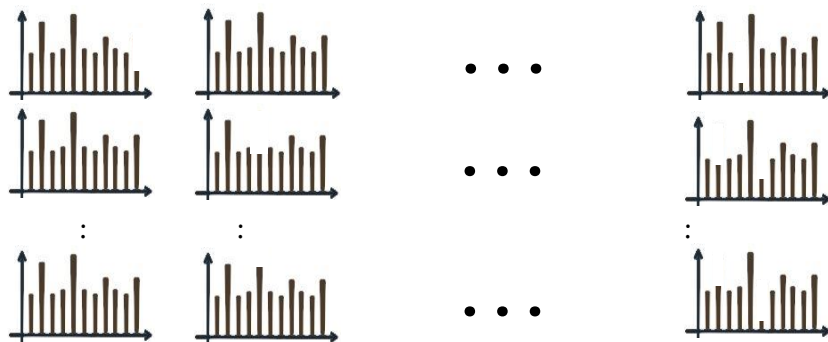


TESTING

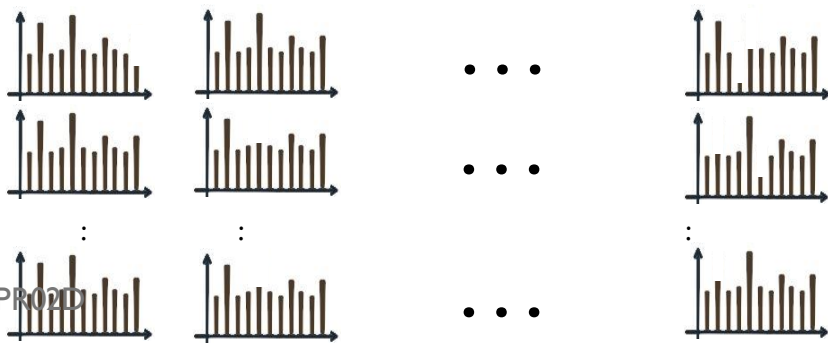


Quién es esta Persona?

Persona 20



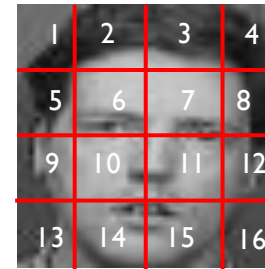
Persona 40



Persona 1

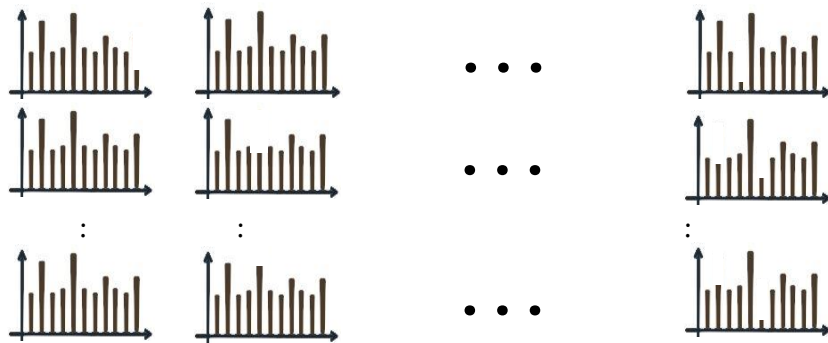


TESTING

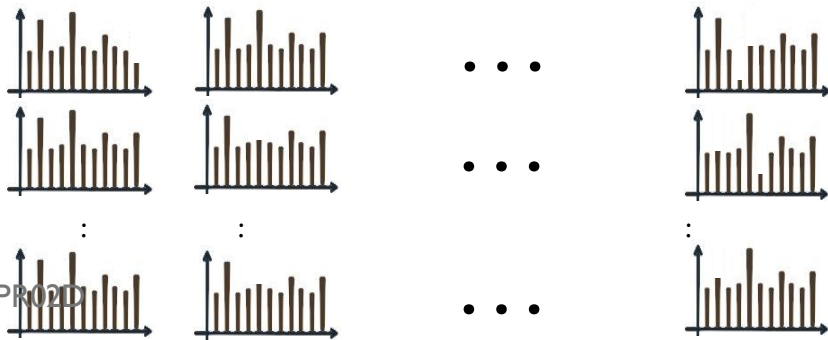


Quién es esta Persona?

Persona 20



Persona 40



Persona 1



Δ

920

TESTING



Quién es esta Persona?

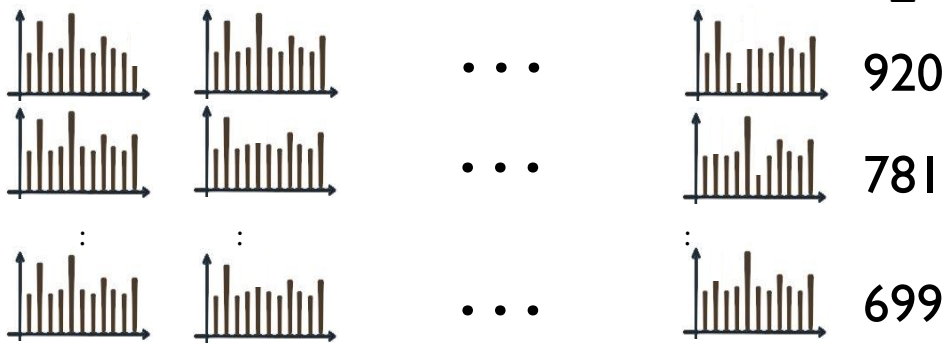
Persona 20



Persona 40



Persona 1



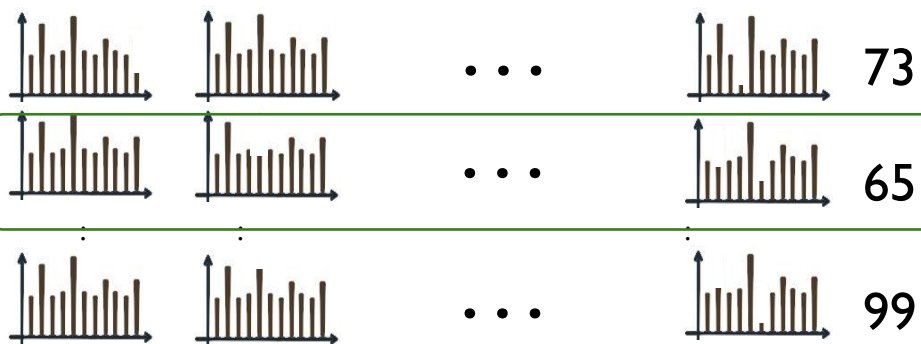
Δ

TESTING



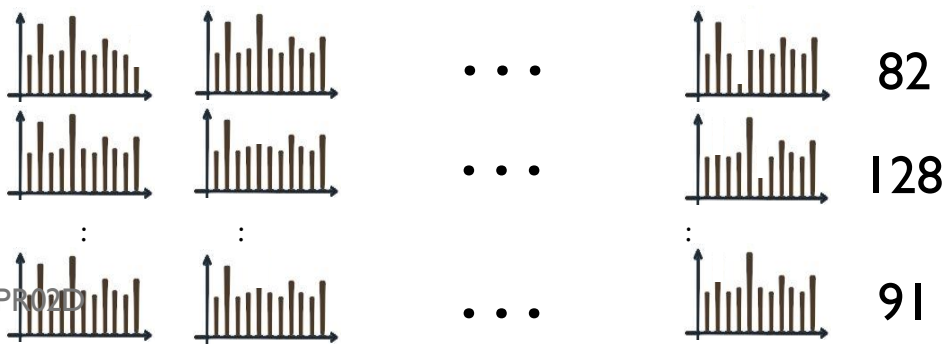
Quién es esta Persona?

Persona 20



mínimo

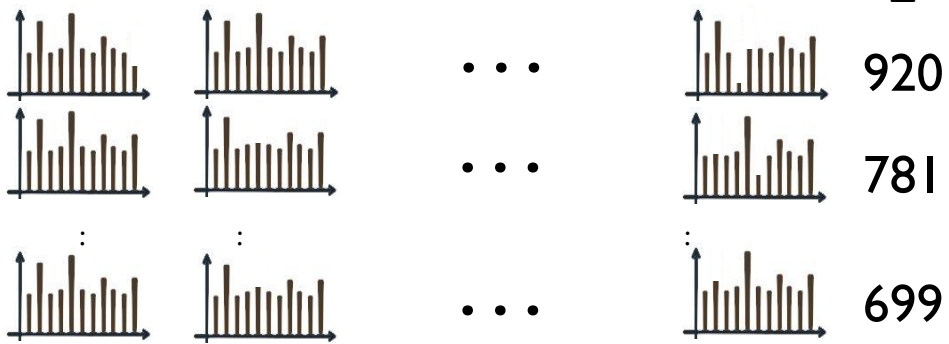
Persona 40



Estrategia I:
El vecino más cercano

Estrategia 2:
k – vecinos más cercanos
(knn)

Persona 1

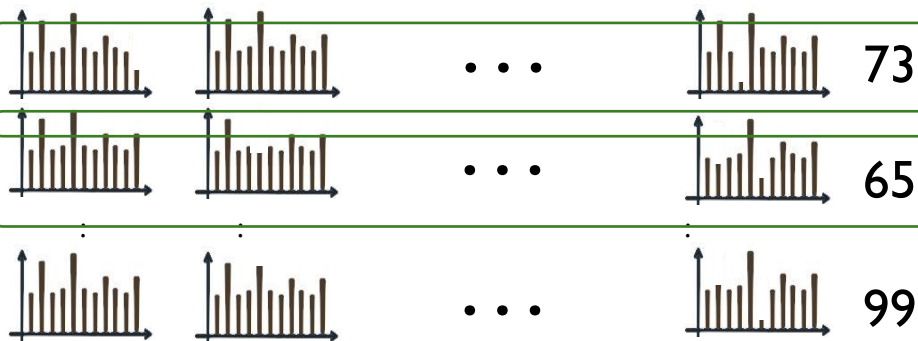


TESTING



Quién es esta Persona?

Persona 20



2nd mínimo

1st mínimo

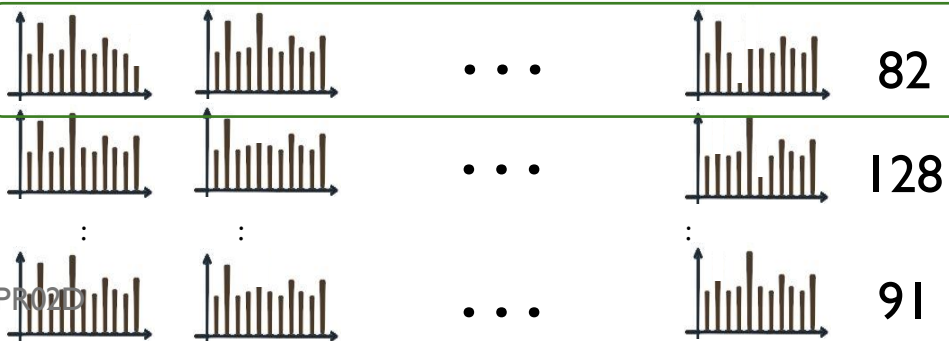
>

>

Persona 20

Persona 20

Persona 40



3rd mínimo

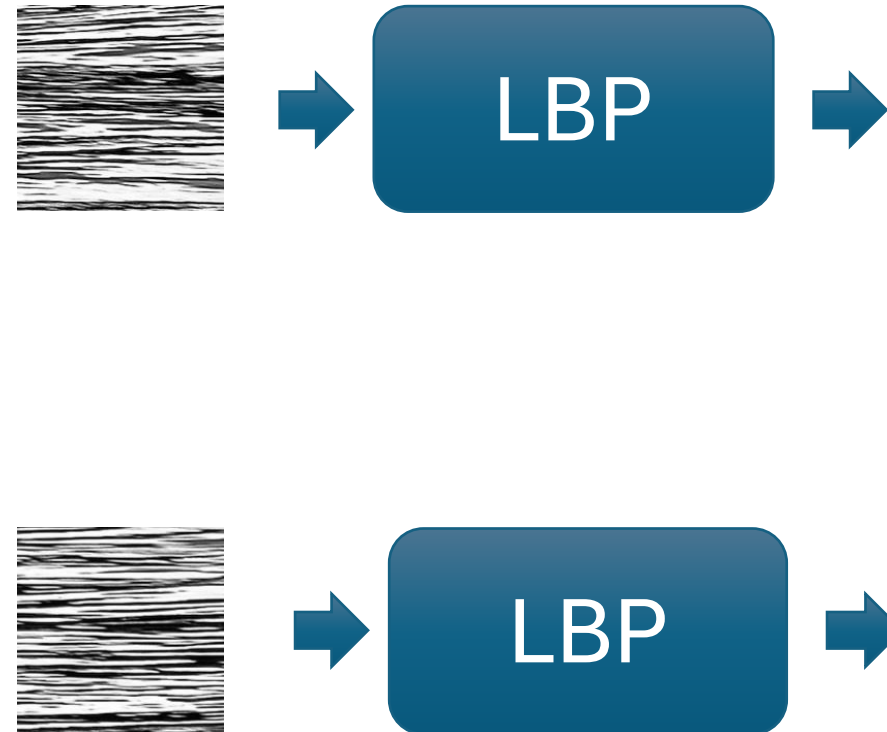
>

Persona 40

Estrategia 2:
k-vecino más cercano (k=3)

LBP en la práctica (alternativa)

FOTOS DE LA MISMA TEXTURA



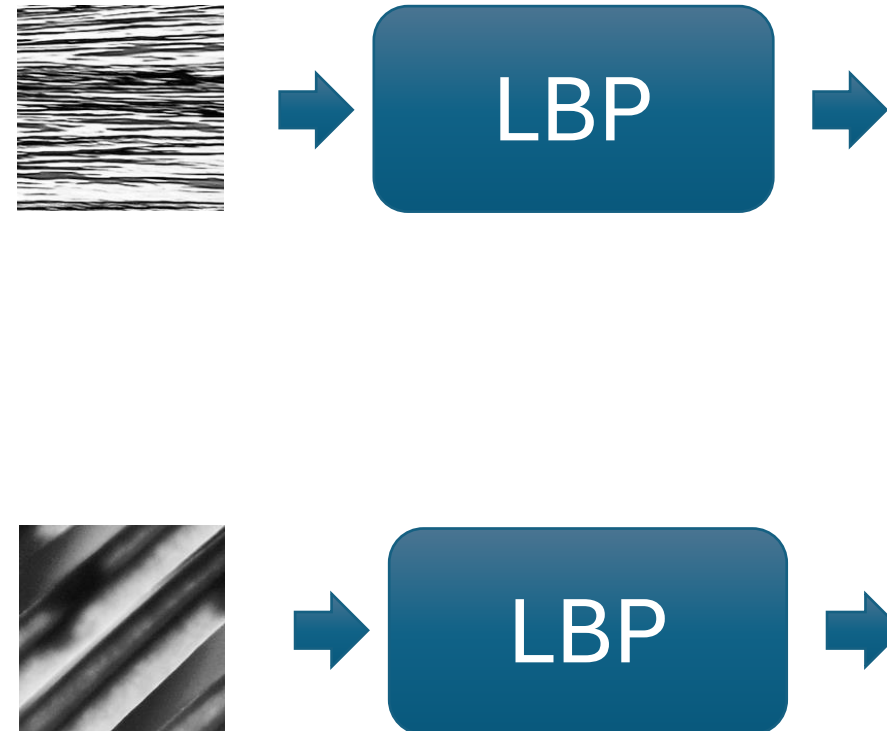
SCORE:

mide qué tan parecidos
son los descriptores



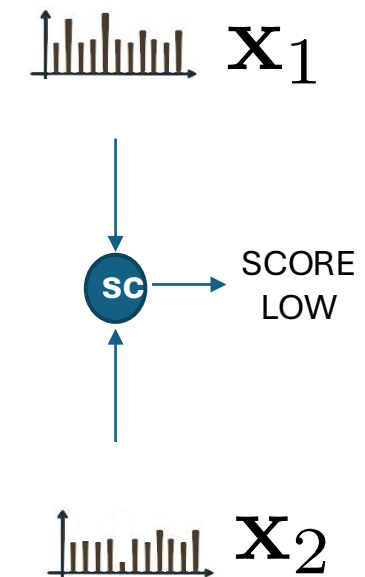
LBP en la práctica (alternativa)

FOTOS DE TEXTURAS DISTINTAS



SCORE:

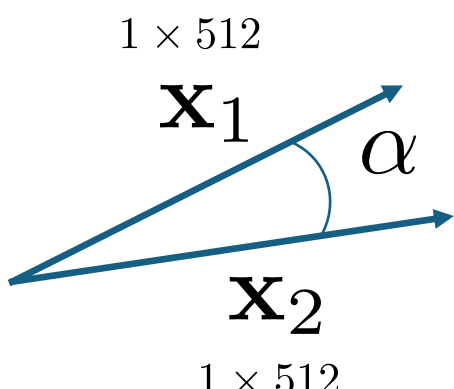
mide qué tan parecidos
son los descriptores



¿QUÉ TAN PARECIDOS SON DOS VECTORES?

NO OLVIDAR: en nuestro caso los vectores tienen norma 1

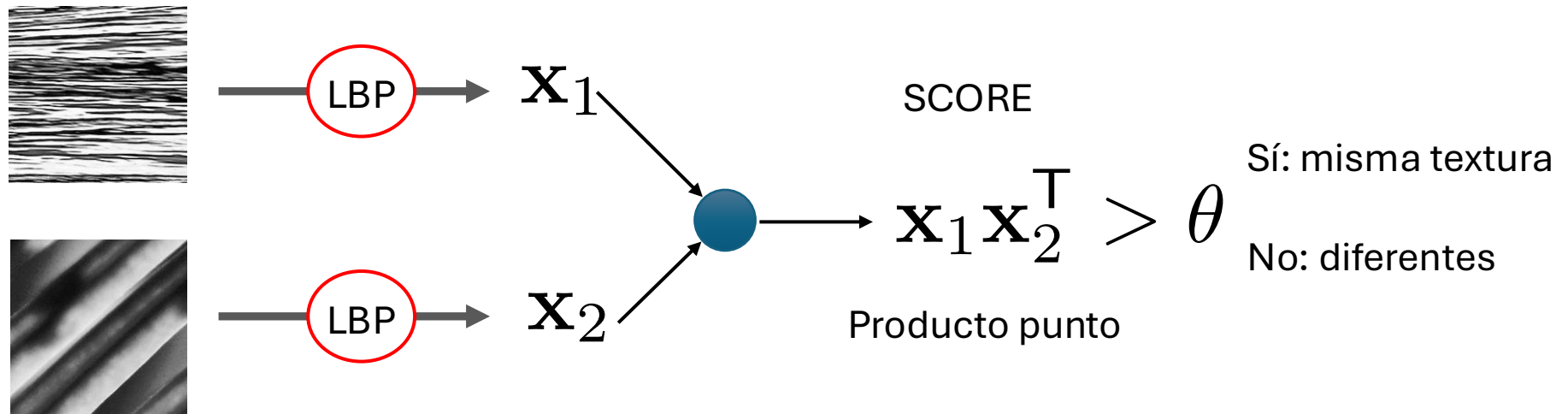
PRODUCTO PUNTO


$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cancel{||\mathbf{x}_1||} \cancel{||\mathbf{x}_2||} \cos(\alpha)$$
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cos(\alpha) \quad \text{SCORE}$$
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \mathbf{x}_1 \mathbf{x}_2^T$$

- Si los vectores son muy parecidos, el ángulo entre ellos es pequeño y el SCORE es cercano a 1.
- Si los vectores son muy distintos, el ángulo entre ellos es grande y el SCORE disminuye significativamente.

LBP en la práctica (alternativa)

Fotos de la misma (diferente) textura tienen descriptores similares (diferentes)



Estrategia 3:
Distancia de clase muestral
más pequeña

Persona 1

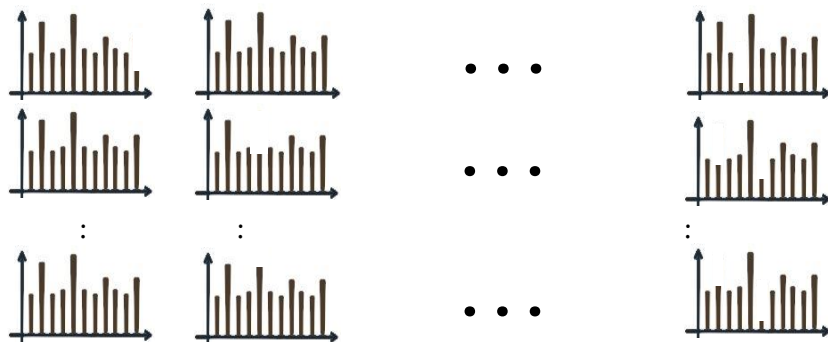


TESTING



Who is this Persona?

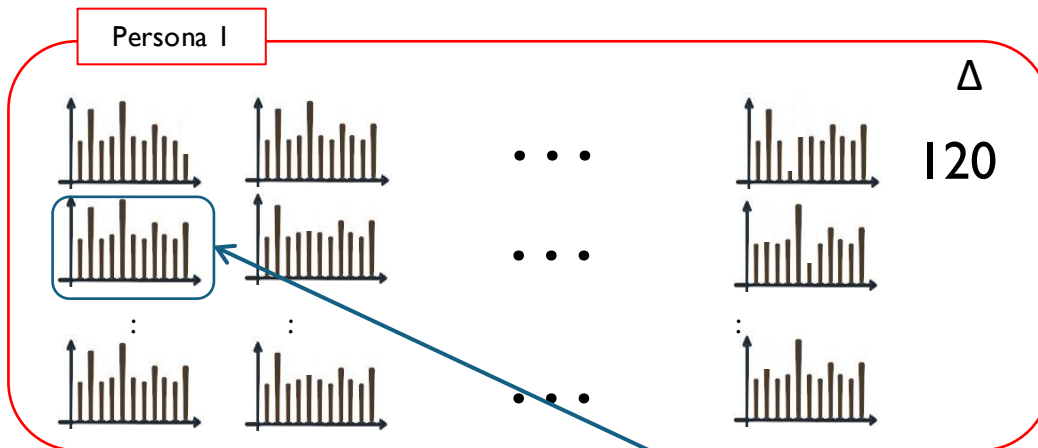
Persona 20



Minimal distance

Persona 40

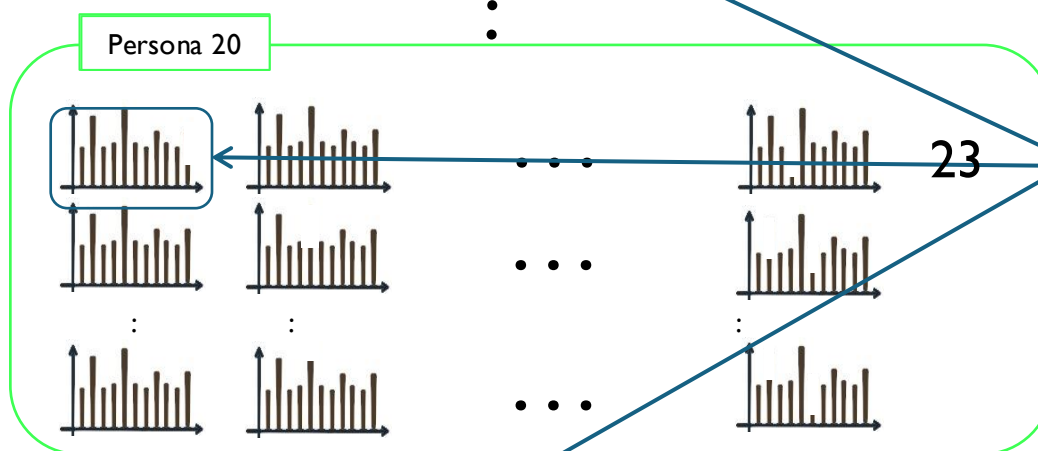




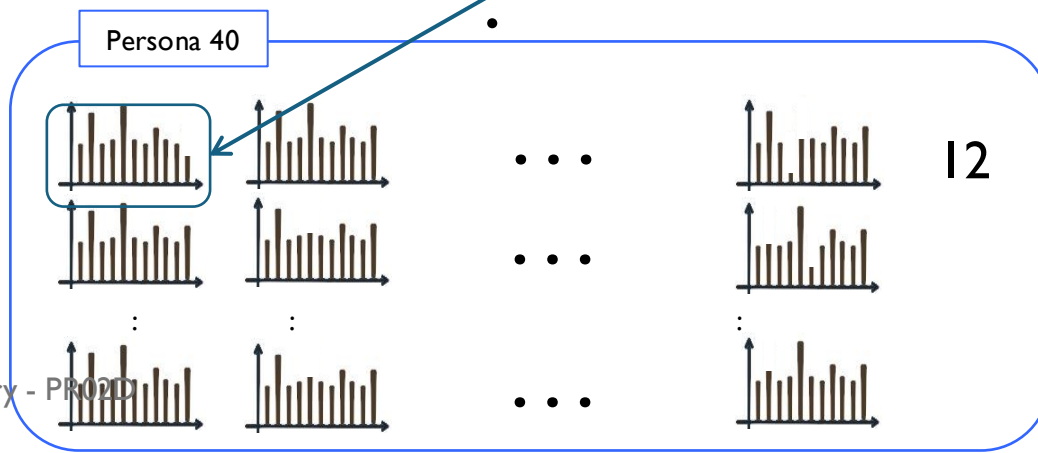
TESTING

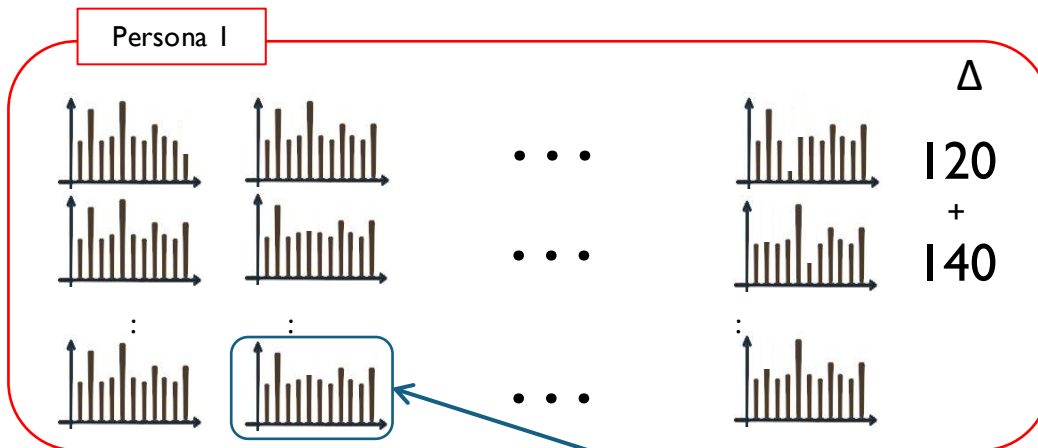


Who is this Persona?



Minimal distance

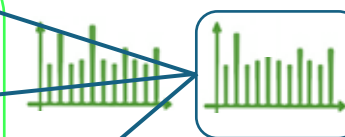
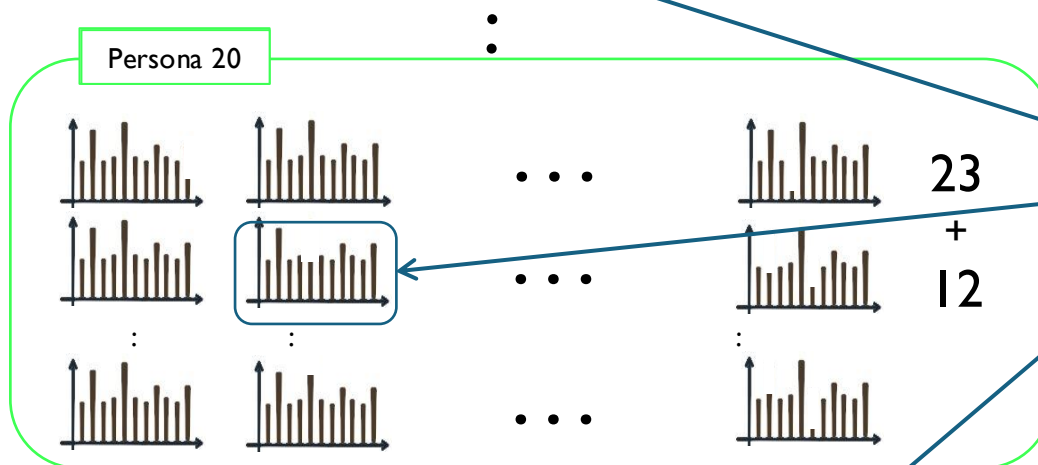




TESTING



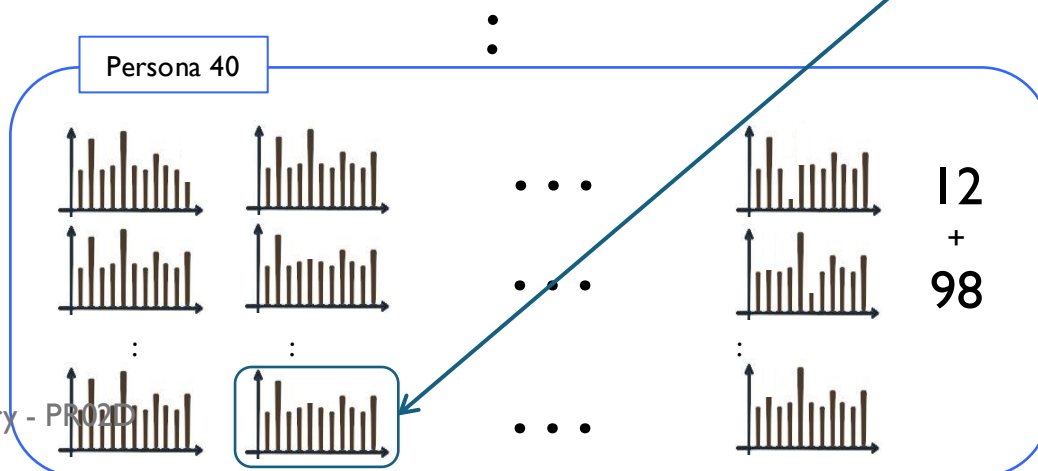
Who is this Persona?

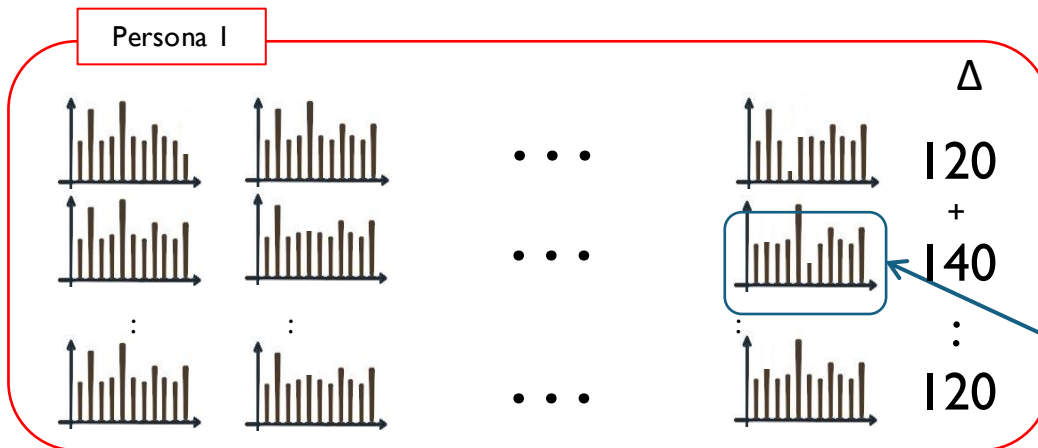


...



Minimal distance

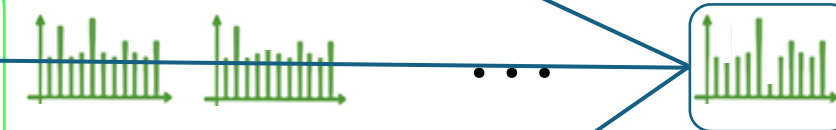
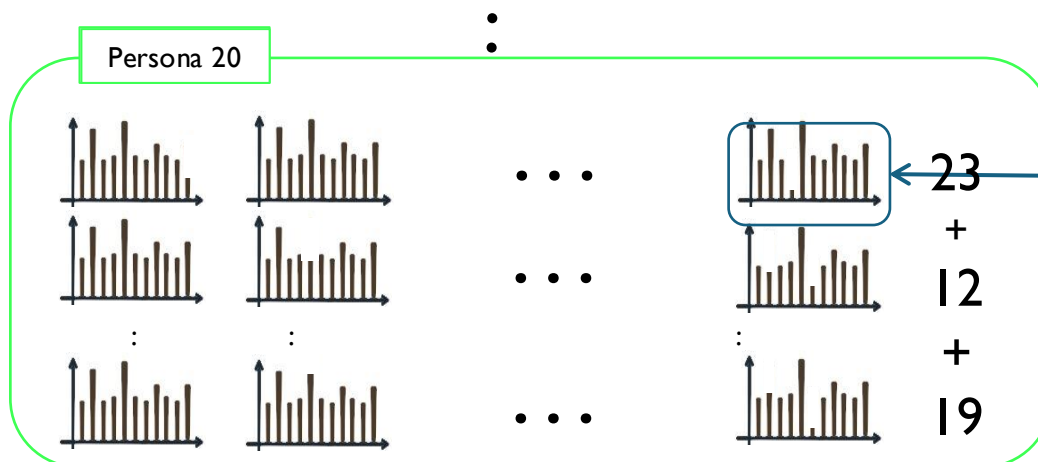




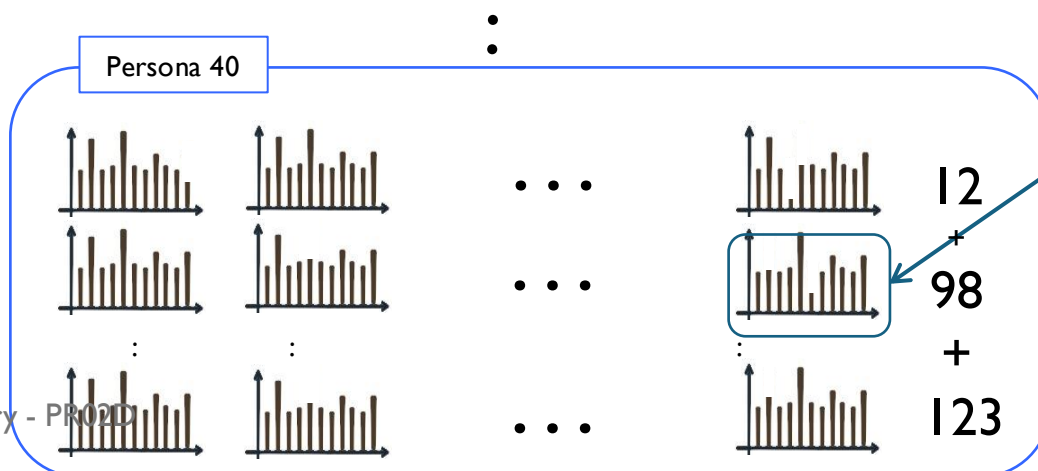
TESTING



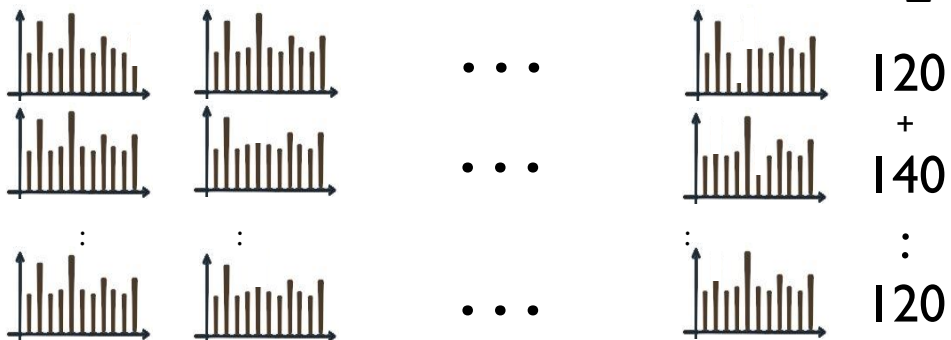
Who is this Persona?



Minimal distance



Persona 1



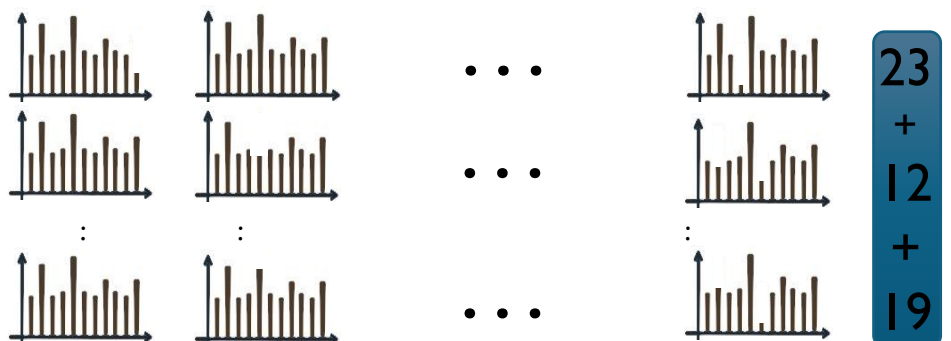
Δ

TESTING



Who is this Persona?

Persona 20

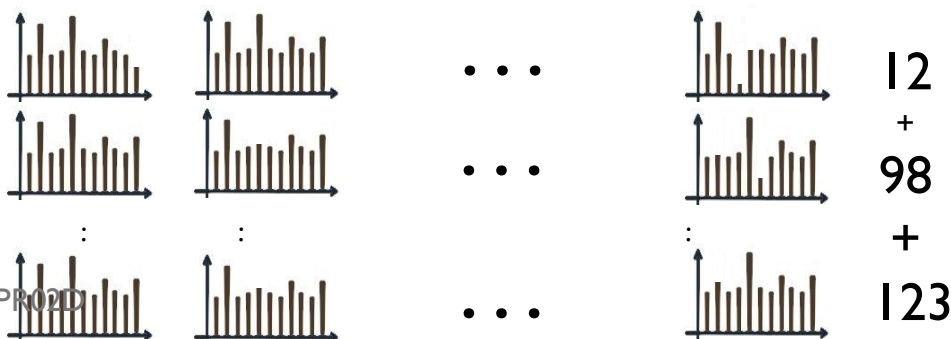


23
+
12
+
19

= 54

Minimal total distance

Persona 40



¿Cómo hacer un LBP invariante a la rotación?

(LBP-ri)

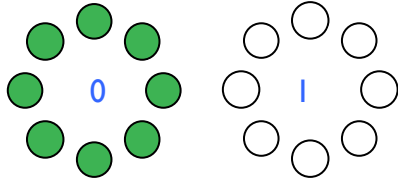
Solución: se codifican todas los patrones del mismo número de activaciones (de todas las direcciones) con un solo código

Patrones uniformes

$2 + 56 = 58$
patrones

$U = 0$

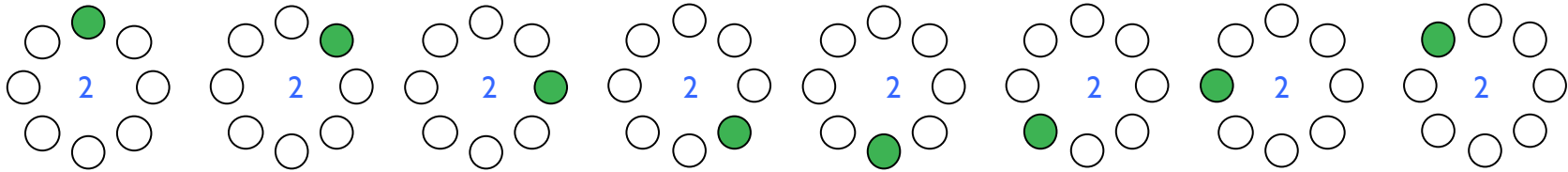
2
patrones
 $\{0, 1\}$



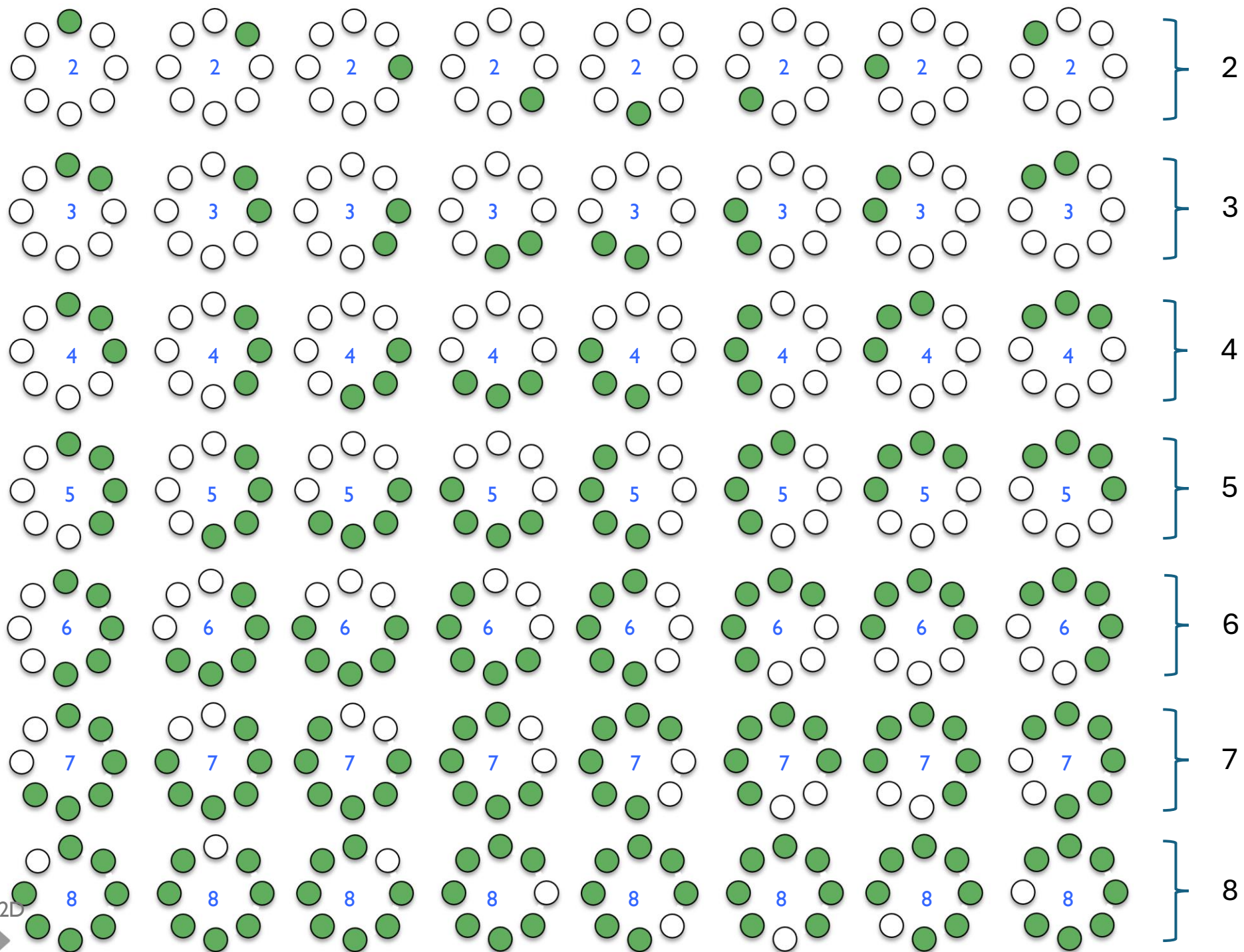
$U = 2$

$8 \times 7 = 56$
patrones

$\{2, 3, \dots, 9\}$



$U=2$



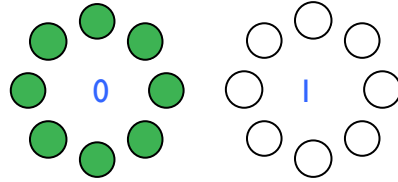
Patrones uniformes

$2 + 56 = 58$
patrones

$U = 0$

2
patrones

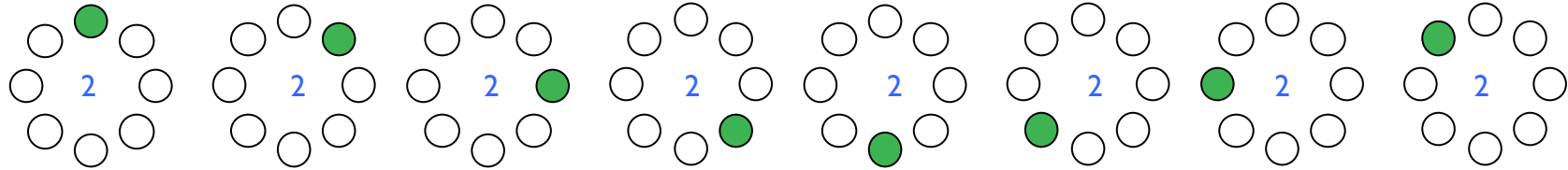
$\{0, 1\}$



$U = 2$

$8 \times 7 = 56$
patrones

$\{2, 3, \dots, 9\}$



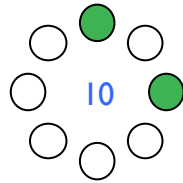
⋮

Patrones no uniformes

$256 - 58 = 198$
patrones

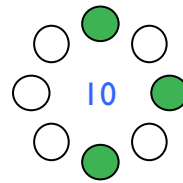
$\{10\}$

$U = 4$



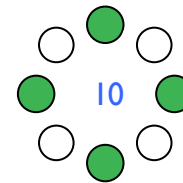
⋮

$U = 6$



⋮

$U = 8$



⋮